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**ANALYSIS AND SIMPLIFICATION OF PRIVACY POLICY
DOCUMENTS AT SCALE**

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by
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Abstract

A privacy policy is a legal document that organisations use to disclose the ways in which they collect, use, share and secure their users’ personal data. Various laws around the world such as General Data Protection Regulation (GDPR) make it mandatory for organisations to make their privacy policies readily available to their users. These laws work under the assumption that users will read the privacy policy of an organisation and either accept the practices or abstain from using the offered services. However, studies have shown that average internet users rarely read privacy policies as they are either too long or too complicated to understand. Moreover, suggestions to improve the comprehensibility of privacy policies have not been adopted by most organisations.

Natural language processing (NLP) can be used to ease the burden of reading privacy policies by extracting salient details from them. However, privacy policies are usually written in legalese and require legal expertise for accurate interpretation. This makes the process of annotating privacy policies expensive, thereby restricting the use of supervised learning approaches. Current research has taken several steps to simplify privacy policies but has had limited success in terms of accuracy, generalizability and real-world application. As such, we follow a multi-faceted approach to improving the online user privacy landscape. We enable users to make informed privacy decisions, lower the barrier for privacy researchers to study policies, and empower policy regulators to enforce compliance of a vast number of constantly updated policies.

In this work, we first create a large-scale longitudinal corpus of privacy policies and train a privacy policy language model to improve the performance of various downstream NLP tasks in the domain. We then evaluate the language model on question answering and text classification tasks. Next, we analyze the availability of privacy policies to internet users, and analyze the various failure modes of privacy policy availability to estimate a lower bound of the total number of privacy policy documents in the world. Next, we create a search engine, which acts as a resource to build new custom corpora and compare data practices between policies. Finally, we create privacy-enhancing technologies that aid users in understanding data practices detailed in policies. With these tools and techniques, we aid in disentangling the barriers to privacy research and improve user comprehensibility of privacy policies.

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Chapter 1 —

Introduction

A privacy policy is a legal document that organisations use to disclose to its users the ways in which their personal data is collected, used, shared and secured online. Personal data can refer to a number features that could be used to identify an individual. Legal jurisdictions around the world require organisation to make their privacy policies readily available to its users through laws like General Data Protection Regulation (GRDP) in the European Union, California Consumer Privacy Act (CCPA) and Children’s Online Privacy Protection Act (COPPA) in the United States. These laws work under the principle of the ‘Notice and Choice’ framework (L. F. Cranor, 2012). According to the notice and choice paradigm, organisations make their privacy practices readily available to its users thereby giving them a ‘notice’ of their privacy practices, and users then have the ‘choice’ of either accepting the privacy practices or abstaining from using the offered services (Sloan & Warner, 2014).

Although the notice and choice framework is theoretically reasonable, there is some evidence that it does not work in practice. Various studies have shown that although users care about their online privacy, they either do not read privacy policies (Rudolph et al., 2018) or often struggle to understand their contents (Obar & Oeldorf-Hirsch, 2018). It has been shown that users would need to spend a substantial amount of time to read all the privacy policies that they come across. McDonald et al. estimate that it would take approximately 200 hours to read all the privacy policies that an average person would come across every year (McDonald & Cranor, 2008). This situation can be expected to get worse as users come across greater number of online services every year. Moreover, privacy policies are hard to comprehend due to the use of legalese (Meiselwitz, 2013). Studies calculating the readability of privacy policies using standard readability measures have shown that privacy policies are hard to comprehend and are at a college reading level (Ermakova et al., 2015).

Various studies have suggested methods to improve the comprehensibility of privacy policies by suggesting improvement in their presentation, but these suggestions have not been adopted by organisations. For example, a ‘privacy nutrition label’ approach to present information in privacy policies suggested that users reported higher speed, accuracy and enjoyment in finding privacy information from a privacy nutrition label rather than a privacy policy (Kelley et al., 2010, 2009). Similarly, privacy policies with machine readable formats failed due to lack of adoption (L. Cranor, 2002). Privacy short notices (Zimmeck & Bellovin, 2014) and suggestions of privacy icons supported by the European Union have not gained traction as well.

User burden in reading privacy policies can be eased using machine learning and natural language processing (NLP) techniques. NLP techniques to simplify privacy policies tested on small corpora of policies (Wilson et al., 2016) (Zimmeck & Bellovin, 2014) (Zaeem et al., 2018) have shown promising results. Several NLP tasks such as privacy practice classification (Wilson et al., 2016), privacy question answering (Ravichander et al., 2019), vague sentence detection (Lebanoff & Liu, 2018), and detection of compliance issues (Zimmeck et al., 2019) have been used to improve the comprehensibility of policies. However, they face issues of accuracy, scalability and generalization due to their small datasets, often consisting of fewer than 10K policies. This paucity of large datasets leads to a lack of robust techniques which could be used to easily understand the wide range of privacy policies on the web. This puts an onus on researchers, policy regulators and legal experts to protect users from unfair privacy practices. However, they are not well equipped to regulate privacy policies at scale. Regulators in some jurisdictions rely on user complaints to investigate privacy practices (*Factsheet 5 - What to expect when we inspect*, 2018) while others rely on organisations to self-certify their compliance ¹.

Summarizing the existing issues, we find that internet users often do not read privacy policies as they are too long and complicated to understand. NLP techniques applied to simplify privacy policies have achieved some success but face limitations in scalability and generalisations due to the lack of corpora of privacy policies. Although this shifts the burden of protecting users from unfair privacy practices on privacy regulators, researchers and experts, they face a similar scalability issue.

In order to address the above mentioned issues, we follow a multi-faceted approach to improving the online user privacy landscape. We enable users to make informed privacy decisions, lower the barrier for privacy researchers to study policies, and empower policy regulators to enforce compliance of a vast number of constantly updated policies.

¹<https://www.privacyshield.gov/Program-Overview>

We created the largest collection of longitudinal website privacy policies with over 4 million policies over four years and made it readily available for research purposes. A large corpus of website privacy policies will stimulate NLP research and resolve some of the issues with scalability and generalizability.

We created a privacy policy search engine which can be used to search for policies on the web. A privacy policy search engine will give researchers the ability to isolate policies based on key features in order to discover trends in privacy practices online. Similarly, policy regulators and legal experts could use the search engine to find anomalies in policies thereby empowering them to protect user from privacy eroding practices.

Next, we investigate the lack of availability of policies by studying the frequency of various failure modes in privacy policy availability, due to issues such as broken links, empty content, placeholder text and language issues. Our analysis of trends in failure modes of privacy policies throw light on the issues in online privacy landscape thereby motivating further study. Lack of availability of privacy policies is especially worrying since it signifies a breakdown of notice and choice and therefore warrants careful investigation. Next, we undertake the task of estimating the lower bound on the total number of privacy policies on the web using the capture-recapture (Lincoln-Petersen estimator) technique (Begon et al., 1979) in order to study the trends in distribution of policies on the internet. We estimate the distribution of website privacy policies in terms of sectors of commerce and the most popular top level domains (TLDs).

Next, we concentrate on building some privacy enhancing technologies (PETs). We present an automated approach to scale identification of data practices in privacy policies, providing users with information regarding their data privacy. By automating the process of aligning data practices with policy excerpts, we take a crucial step towards eliminating the need for users to laboriously parse through lengthy and intricate documents, ensuring that they receive precise and easily digestible information about their online privacy. This lowers the barrier for internet users to understand what is happening with their data online, thereby allowing them to make informed privacy decisions. Next, we create a pipeline to extract policy dates to ease the process of finding valid and up-to-date policies. We extract text from the privacy policy HTML, followed by extracting all date instances and classifying candidate policy dates into either policy date, i.e. updated date or effective date, or other types of dates using a transformer-based language model. We investigate trends in updating privacy policies and find that the dates when new privacy laws became effective have had a significant impact on when most privacy policies have been updated. Our findings strongly suggest that a large percentage of

policies online are not up-to-date and are not compliant with legal requirements.

Finally, we discuss the implications of this work and directions for future work. The rest of the document is arranged as follows: Chapter 2 provides a brief overview of the research on user privacy needs and the state-of-the-art for natural natural language processing tasks in the privacy domain. Chapter 3 focuses on PrivBERT, the privacy policy language model and the data pipeline used to gather a large corpus of policies. Chapter 4 discusses the availability of privacy policy documents. Chapter 5 discusses the search engine in details. Chapter 6 discusses privacy enhancing technologies that we built. Finally, Chapter 7 explains the socio-technical implications of this research and provides direction for future work.

Chapter 2 —

Related Work

2.1 User Privacy

Studying user privacy has had a long history. Early experiments studying user concern for online privacy in the United States showed that 58% of users were very concerned about their privacy online, while 27% were somewhat concerned (O’Neil, 2001). Other studies zeroed-in on user privacy preferences in various sectors of online activity (Ackerman et al., 1999; Spiekermann et al., 2001). These studies divided internet users into three categories - privacy fundamentalists who were extremely concerned about their online privacy, the pragmatic majority who are concerned about their privacy but could be appeased if specific concerns were met by policies, and finally the marginally concerned who didn’t care about online privacy and were willing to provide personal information. Earp & Baumer (2003) found that users were willing to provide personal information based on who was asking for it. Earp et al. (2005) examined user perception of privacy risk in relation to online privacy policies and found that there was a lack of alignment in the information that user wanted and information that privacy policies provided. Further studies showed that privacy policies tended to intensify users privacy concerns rather than promote trust (Pollach, 2007).

While studies generally showed that users were very concerned about their online privacy and that privacy policies were not effectively addressing privacy concerns, Barnes (2006) and Norberg et al. (2007) showed that there was a paradox between user concerns and their actions. They showed that while users reported that they were concerned about their online privacy, they were willing to give away their personal information. A vast array of studies have attempted to resolve this paradox. Solutions to the paradox usually found that users had flawed model to calculate costs vs benefits (Barth & De Jong, 2017). In addition, due to issues such as necessity for immediate gratification (Deuker,

2009), users failed to effectively assess costs due to knowledge deficiency or insufficient knowledge (Flender & Müller, 2012; Phelan et al., 2016; Young & Quan-Haase, 2013; Lutz & Strathoff, 2014).

Further studies attempted to understand what motivates users to share their personal information (Leon et al., 2013) and found that the type of information is an important factor to users. Armed with the knowledge of user privacy preferences, tools were implemented which would indicate to users the privacy rating of a website. Privacy Bird (Byers et al., 2004) was one such tool which worked in conjunction and P3P (Platform for Privacy Preferences) and the Internet Explorer. Privacy Bird would display a different colored bird in the search results of a search engine based on the privacy practices of the website. User studies on Privacy Bird found that users who adopted privacy bird improved their privacy practices by reading privacy policies and taking steps to protect their personal information (L. F. Cranor et al., 2002; Vu et al., 2010). More recently, the TOS;DR (Terms of Service; Didn't Read)¹ group started an initiative to annotate privacy policies and terms of service documents. They collect annotations of various privacy practices from internet users and either accept or reject the provided annotation. They also provide a browser extension which a user can use to view the privacy practice of a website.

Responding to the opaqueness of privacy policies, Schaub et al. (2015) introduced methods to ease the design of privacy notices and their integration, and Kelley et al. (2010) designed and tested a *privacy nutrition label* approach to present privacy information visually. Kelley et al. (2013) designed and displayed a short *Privacy Facts* display which showed users what information is collected by Android Apps when they are downloaded. The authors found that by bringing the users privacy information in a clear fashion when they were making the decision of downloading an application, they could assist users in choosing application that requested access to fewer permissions. Similarly, Mugan et al. (2011) used a machine learning approach to learn user privacy preferences and build personas. They found that most users converge into personas with acceptable levels of privacy. Similarly, Lin et al. (2014) found that it was possible to identify a small number of privacy profiles that collectively do a good job at capturing the diverse preferences of users. However, suggestions to improve the presentation of privacy information, have not been adopted by many organisations. Apple has begun displaying privacy labels in its app stores having collected the information from App developers; however, concise privacy information for websites remains an open problem.

¹<https://tosdr.org/>

This body of literature shows that even though users privacy needs and preferences are diverse, they coalesce into various categories. Users preferences can thus be modelled in order to build tools from which users will derive utility. Lack of user behaviour supporting the notion that they cared about their privacy is explained by the need for instant gratification, meaning that the users did not appreciate the overhead necessary to understand privacy practices of organisations. It can also be explained by a lack of knowledge required to weight the benefits against the risk to privacy. This lack of knowledge can be explained once again by the overhead required to understand privacy practice. Therefore an natural language processing based approach which would decrease the overhead to understand privacy practices of organisations should be well received by users.

2.2 Natural Language Processing on Legal Text

Natural Language Processing (NLP) has increasingly been applied to the legal domain, offering valuable tools for automating tasks traditionally reliant on human expertise. Since privacy policies are legal documents, we review various NLP approaches specifically tailored for legal language.

Legal Named Entity Recognition is one of the fundamental task in the legal field, aimed at identifying and categorizing entities within legal documents. These entities often include parties involved in a case (such as appellants and respondents), legal representatives, and judicial authorities. Notable contributions to this area include the dataset developed by Leitner et al. (2020), which focuses on German legal documents, and the work by Kalamkar et al. (2022), which addresses named entity recognition in Indian court judgments.

Legal Summarization seeks to distill lengthy case documents into concise summaries, either through extractive methods, which select key sentences, or abstractive methods, which generate human-like summaries. Research such as that by Shukla et al. (2022) examines both extractive and abstractive approaches for summarizing legal cases, while Sharma et al. (2019) present BIGPATENT, a large-scale dataset aimed at facilitating coherent summarization of legal texts.

Legal Judgment Prediction involves forecasting the outcomes of court cases, a complex task often comprising multiple sub-tasks. Mumcuoğlu et al. (2021) explore outcome prediction in Turkey’s higher courts, while Malik et al. (2021) provide insights through the Indian Legal Documents Corpus for Court Judgment Prediction and Ex-

planation. The CAIL2018 dataset is another significant resource, offering a large-scale dataset specifically for judgment prediction tasks (Xiao et al., 2018).

Pre-trained Language Models for the Legal Domain have gained traction, with efforts to adapt large transformer-based models for various legal tasks. Notable models include LEGAL-BERT (Chalkidis et al., 2020), which has been specifically tailored for legal applications, and research by Zheng et al. (2021), which assesses the benefits of self-supervised learning in legal contexts. General Resources and Benchmarks provide additional frameworks for understanding legal language. The LexGLUE benchmark dataset (Chalkidis et al., 2022) is particularly noteworthy, as it standardizes evaluation across various legal language tasks.

The application of NLP in the legal domain is a rapidly evolving field, with diverse resources and methodologies being developed to enhance the efficiency and effectiveness of legal document processing. These advancements not only support legal professionals but also contribute to a broader understanding of how language technologies can address the complexities of legal language.

2.3 Natural Language Processing and Privacy Policies

2.3.1 Privacy Policy Corpora

The application of NLP approaches to analyse and simplify privacy policies can decrease the burden of reading them for internet users. As such there have been a number of studies that have attempted to apply NLP approaches to simplify privacy policies. Most studies have used small corpora of privacy policies. Wilson et al. (2016) released the OPP-115 Corpus, a publicly available dataset of 115 privacy policies with manual annotations of 23k fine-grained data practices, and demonstrated the feasibility of partly automating the annotation process. The OPP-115 Corpus was used to create Polysis (Harkous et al., 2018), a system that uses deep learning to classify text in privacy policies and answer non-factoid questions. Similarly, the OPP-115 Corpus was used to train ten different machine learning models to summarise 400 privacy policies and answer predefined questions (Zaeem et al., 2018). The OPP-115 Corpus was further used to evaluate embeddings created from 150,000 Android Google Play Store app privacy policies (V. B. Kumar et al., 2019). Researchers Ramanath et al. (2014) released a corpus of 1010 website privacy policies and introduced an unsupervised approach to aligning privacy policy sections. Zimmeck et al. (2019) released a set of over 400k URLs

to Android app privacy policy pages collected by crawling the Google Play store. Their analysis on the policies include classification of various privacy practices and analysing potential compliance issues on 350 annotated policies. Amos et al. (2020) collected privacy policies from around 130,000 websites from over two decades and analysed the evolution of the online privacy landscape. Finally, Nokhbeh Zaeem & Barber (2021) collected a corpus of around 100k privacy policies using the domains from DMOZ, a website which maintained categories of websites on the internet.

Among the above corpora of policies, only a couple have specifically focused on making privacy policies more accessible to the public. The Usable Privacy Policy Project made available the OPP-115 Corpus (Wilson et al., 2016). These policies can be accessed through the website and filtered based on the URL². They display the sector of activity, readability and popularity of the website from which the policy was originally obtained. In addition, they also created a collection of seven thousand privacy policies which contain machine annotated privacy practices which can be filtered by URL. Similarly, Polisis³ is a collection of about 31,000 privacy policies which generates automatic summaries of privacy policies based on various data practices. While the privacy policies can be filtered based on their URL, they cannot be searched based on user queries (Harkous et al., 2018).

2.3.2 Applying NLP for Policy Analysis and Simplification

Some prior attempts at analysis involved manual analysis of privacy policies. Jensen & Potts (2004) evaluate the accessibility, writing, and content of 64 privacy policies. L. F. Cranor et al. (2013) compare the privacy policies of over 3000 financial institutions and report on a range of data collection practices.

A large body of work exists which has attempted to analyse and simplify privacy policies automatically. The earliest approaches to privacy policy classification have been using classical machine learning techniques to classify privacy practices. Costante et al. (2012) tried to assess the completeness of privacy policies based on the existence of one or more privacy practices in the policy. Zimmeck & Bellovin (2014) take it one step further by training a Naive Bayes classifier to automatically predict whether a privacy policy follows six privacy practices. Wilson et al. (2016) created a baseline for classifying privacy policy text into one of ten categories roughly based on privacy practices. Nejad et al. (2020) used BERT vector representations (Devlin et al., 2018) to

²<https://explore.usableprivacy.org/>

³<https://pribot.org/polisis>

enrich the OPP-115 corpus and trained a convolutional neural network for the privacy practice classification task. Harkous et al. (2018) used the OPP-115 Corpus to train models to improve upon the results of classifying privacy practice categories. They also created a chatbot called PriBot which could answer users’ privacy questions about a privacy policy. Approaches to answering users’ privacy questions regarding a privacy policy have been similar to the ones used on reading comprehension. The PrivacyQA corpus contains 1,750 questions and expert-annotated answers for the privacy question answering task (Ravichander et al., 2019).

A popular method of classifying privacy policies is based on the extraction of specific features from privacy policies. Various machine learning methods have been applied to test the compliance of privacy policies. Story et al. (2019) frame the problem of compliance as a classification problem in which they classify if the permissions that a mobile app requests correspond to the privacy practices declared in the policy. The authors use a term-frequency-based vectorization technique to train a support vector machine and achieve F1 scores ranging from 75% to 100%. Similarly ‘KnIGHT’ is a tool which can be used to map excerpts in privacy policies to relevant GDPR articles (Nejad et al., 2018). KnIGHT uses a semantic text matching approach from an information retrieval perspective. F. Liu et al. (2016) use a recurrent neural network based approach to model vagueness of privacy policies. They use pretrained Word2Vec embeddings (Mikolov et al., 2013) to train a model to both predict the next word in a sequence and predict if a given word is vague or not. Along similar lines, Nejad et al. (2019) use a machine learning approach to predict risk factors in privacy policies. They first use a deep learning approach to categorise privacy policies into privacy practices and use a rule based system to extract the values of low level attributes and match them with risk levels. Similarly, Lebanoff & Liu (2018) constructed the first corpus of human-annotated vague words and sentences in privacy policies and studied automatic vagueness detection. Sathyendra et al. (2016) used the OPP-115 Corpus to extract opt-out choices from privacy policies. Similarly, Bannihatti Kumar et al. (2020) created their own opt-out choice corpus with the help of crowdworkers and improved the results previously obtained on the opt-out choice extraction task. They also created a browser extension that users can install into their browser that automatically reads a website’s privacy policy and shows the opt-out choices that a user has.

Chapter 3 —

Privacy Policy Language Modeling

3.1 Background

Natural language processing (NLP) provides an opportunity to automate the extraction of salient details from privacy policies, thereby reducing human effort and enabling the creation of tools for internet users to understand and control their online privacy. Existing research has achieved some success using expert annotated corpora of a few hundred or a few thousand privacy policies (Wilson et al., 2016; Zimmeck et al., 2019; Ramanath et al., 2014), but issues of accuracy, scalability and generalization remain. More importantly, annotations in the privacy policy domain are expensive. Privacy policies are difficult to understand and many tasks such as privacy practice classification (Wilson et al., 2016), privacy question answering (Ravichander et al., 2019), vague sentence detection (Lebanoff & Liu, 2018), and detection of compliance issues (Zimmeck et al., 2019) require skilled legal experts to annotate the dataset. In contrast, approaches involving large amounts of unlabeled privacy policies remain relatively unexplored.

Modern robust language models, such as transformer-based architectures, benefit from increasingly large training sets. These models can be used on downstream tasks (Devlin et al., 2019) to improve performance. Results have shown that in-domain fine tuning of such pre-trained language models have produced a significant boost in performance on many tasks (Gururangan et al., 2020) in a variety of domains, suggesting a need for a larger collection of privacy policies to enable similar results in the privacy domain.

To satisfy the need for a much larger corpus of privacy policies, we introduce the PrivaSeer Corpus of over 4 million English language website privacy policies. Here we describe the first of five crawls to create the PrivaSeer Corpus. Subsequent crawls followed an identical pipeline and added to the total number of privacy policy docu-

ments. In the rest of this chapter, we only refer to the first crawl. We describe the corpus creation pipeline, with stages including a web crawler, language detection, document classification, duplicate and near-duplication removal, and content extraction. We then analyse the lengths and top-level distribution of the privacy policies in the corpus and use topic modelling to explore the component topics. Subsequently, we pre-train PrivBERT, a transformer-based language model, using the corpus and evaluate it on data practice classification and question answering tasks. We release the corpus (Srinath, Sundareswara, et al., 2021), the document collection pipeline, and a language model to support further research in the privacy domain.¹

3.2 Related Work

Prior collections of privacy policy corpora have led to progress in privacy research. Wilson et al. (2016) released the OPP-115 Corpus, a dataset of 115 privacy policies with manual annotations of 23k fine-grained data practices, and they created a baseline for classifying privacy policy text into one of ten categories. The corpus was used to train models to extract opt-out choices from privacy policies (Sathyendra et al., 2016), to automatically identify policies on websites and find compliance issues (Story et al., 2019), and to classify privacy practices and answer privacy related non-factoid questions (Harkous et al., 2018).

Other corpora similar to OPP-115 Corpus have enabled research on privacy practices. The PrivacyQA corpus contains 1,750 questions and expert-annotated answers for the privacy question answering task (Ravichander et al., 2019). Similarly, Lebanoff & Liu (2018) constructed the first corpus of human-annotated vague words and sentences in privacy policies and studied automatic vagueness detection. Sathyendra et al. (2017) presented a dataset and developed a model to automatically identify and label opt-out choices offered in privacy policies. Similarly, Zimmeck et al. (2019) released a set of over 400k URLs to Android app privacy policy pages collected by crawling the Google Play store. Amos et al. (2020) collected privacy policies from around 130,000 websites from over two decades and analysed the evolution of the online privacy landscape. Finally, Nokhbeh Zaeem & Barber (2021) collected a corpus of around 100k privacy policies using the domains from DMOZ, a website which maintained categories of websites on the internet.

¹All artifacts are available at <https://privaseer.ist.psu.edu/>.

3.3 Document Collection

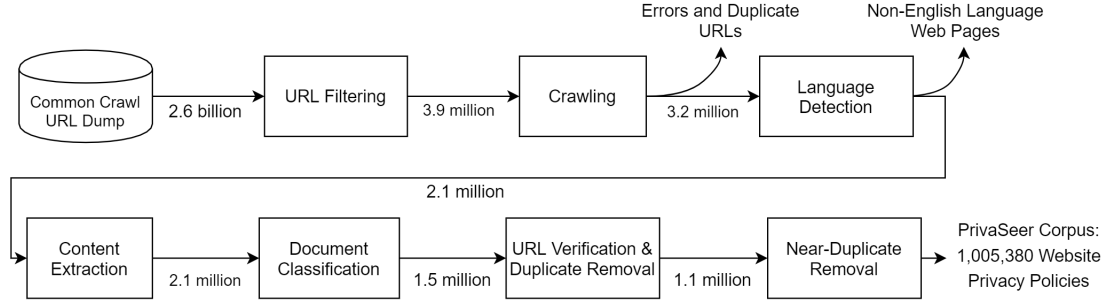


Figure 3.1: Corpus creation pipeline

To build the PrivaSeer corpus, we create a pipeline concentrating on focused crawling (Chakrabarti et al., 1999; Diligenti et al., 2000) of privacy policy documents. We used Common Crawl,² described below, to gather seed URLs to privacy policies on the web. We filtered the Common Crawl URLs to gather a set of possible links to web site privacy policies. We then crawled the filtered set to obtain candidate privacy policy documents. The complete pipeline from the Common Crawl URL dump to the gold standard privacy policy corpus is in Figure 3.1.

3.3.1 Common Crawl

The Common Crawl Foundation has been releasing large monthly internet web crawls along with their web graphs since 2008. Monthly crawl archives provide a “snapshot of the web” by including re-crawls of popular domains and crawls of new domains. We downloaded the URL dump of the May 2019 archive.³ Common Crawl reports that the archive contains 2.65 billion web pages or 220 TB of uncompressed content which were crawled between 19th and 27th of May, 2019. We applied a selection criteria on the downloaded URL dump to filter the URLs of likely privacy policy pages.

3.3.2 URL Selection

Due to legal requirements, organizations typically include a link to their privacy policy in the footer of the website landing page commonly with the names *Privacy Policy*, *Privacy Notice*, and *Data Protection*. We selected those URLs which had the word “privacy”

²<https://commoncrawl.org/>

³<https://commoncrawl.s3.amazonaws.com/crawl-data/CC-MAIN-2019-22/cc-index.paths.gz>

or the words “data” and “protection” from the Common Crawl URL archive. We were able to extract 3.9 million URLs that fit this selection criterion. Informal experiments suggested that this selection of keywords was optimal for retrieving the most privacy policies with as few false positives as possible. To find the accuracy of this technique, we manually examined 115 English language website landing pages and their privacy policy URLs from the OPP-115 Corpus (Wilson et al., 2016) since it was built to cover the diverse distribution of privacy policies on the web, in terms of website popularity and sector of commerce. We found that out of 115 websites, 4 websites did not have their privacy policy links either on the landing page or one hop from the landing page and 5 other websites did not satisfy our URL selection criteria. Thus, our crawling technique would cover about $92.17\% \pm 6.51\%$ of English privacy policies on the web with a 95% confidence interval.

3.3.3 Web Crawling

We crawled the 3.9 million selected URLs using Scrapy⁴ for about 48 hours between the 4th and 10th of August 2019, for a few hours each day. 3.2 million URLs were successfully crawled, henceforth referred to as *candidate* privacy policies, while 0.4 million led to error pages and 0.3 million URLs were discarded as duplicates.

3.4 Document Filtering

3.4.1 Language Detection

We focused on privacy policies written in the English language, to enable comparisons with prior corpora of privacy policies. To identify the natural language of each candidate document, we used the open-source Python package Langid (Lui & Baldwin, 2012a). Langid is a Naive Bayes-based classifier pretrained on 97 different languages, designed to achieve consistently high accuracy over a wide range of languages, domains, and lengths of text. The complete set of documents was divided into 97 languages and an unknown language category. We found that the vast majority of documents were in English. We set aside candidate documents that were not identified as English by Langid and were left with 2.1 million candidates.

⁴<https://scrapy.org/>

3.4.2 Content Extraction

Manual inspection of the English language web pages showed that they included content other than the main text: often they had a header, a footer, a navigation menu, and banners. We refer to this extra content in a web page as *boilerplate*. Boilerplate draws away from the focus of the main content in a web page and therefore various techniques have been used to remove boilerplate from web pages (Gottron, 2007; Weninger et al., 2016). After manual comparison of a number of content extraction tools, we used the open-source Python package *boilerpipe* (Kohlschütter et al., 2010) due to its superior performance. Boilerpipe effectively strips web pages of boilerplate using shallow text features, structural features and density based features.

3.4.3 Document Classification

Some of the web pages in the English language candidate document set may not have been privacy policies and instead simply satisfied our URL selection criteria. To separate privacy policies from other web documents we experimented with supervised and unsupervised approaches.

3.4.3.1 Unsupervised Machine Learning

For the unsupervised approach, we uniformly randomly sampled 100,000 documents and experimented with K-Means (Lloyd, 1982) and DBSCAN (Ester et al., n.d.) algorithms. We tested our models using Doc2Vec (Le & Mikolov, 2014) and term frequency-inverse document frequency (tf-idf) vectorization techniques. The Doc2Vec technique was used to create 256 dimensional vectors and the tf-idf technique created vectors equal to the size of the vocabulary. Both were further reduced to 50 by using Principal Component Analysis (PCA). A cosine distance metric was used to compare the document distances.

For the K-Means algorithm, using Doc2Vec vectors we experimented with cluster sizes between 2 and 30 and could not find a clear elbow point. As manual analysis of documents for various cluster sizes did not reveal an intuitive explanation for formed clusters, we discarded the results. While using the tf-idf vectors, although a clear elbow point was found at 7 clusters, a single cluster was found to contain almost all the documents while the rest of the clusters had between 1 and 1500 documents. As this did not seem like a reasonable separation, we discarded the results.

For the DBSCAN algorithm, using Doc2Vec vectors, over 70% of the data points fell into the noise category for any reasonable range of hyper-parameters. Since this also was

an unreasonable result, we discarded the results. Finally, while using the tf-idf vectors, after hyper-parameter tuning using 320 labelled samples we found that two well defined clusters were formed with 80% of the samples falling under a single category and the other 20% falling under the noise category. The evaluation of this model can be found in Table 3.1.

3.4.3.2 Supervised Machine Learning

Two researchers in the team labeled 1,600 randomly selected candidate documents based on a preset scheme in consultation with a privacy expert. While both the researchers had substantial prior experience with privacy policies, the privacy expert was consulted to eliminate uncertainty in the annotations of a few documents. Lack of agreement in the annotations occurred for six documents, which were settled by discussion with the expert. This occurred in cases when documents that discussed privacy practices were either very short or occurred together with another policy, such as a cookie policy or a terms and conditions document. Out of 1,600 documents, 1,145 were privacy policies and 455 were not privacy policies. Out of the documents that did not have privacy policies, 112 had news articles, 34 were pages which had links to the privacy policy, 27 were e-commerce product pages, 19 were pages which advertised security devices, 12 were Twitter pages and the rest were other miscellaneous pages.

We trained four supervised machine learning models using the manually labelled documents with features extracted from the URLs and the words in the web page. We trained three random forest models and fine-tuned a transformer based pretrained language model, namely RoBERTa (Y. Liu et al., 2019). The three random forest models were trained on three different sets of features: one using the features extracted from the URL, one using the features extracted from the document content, and a combined model using features from both.

For the URL model, the words in the URL path were extracted and the tf-idf of each term was recorded to create the features (Baykan et al., 2009). As privacy policy URLs tend to be shorter and have fewer path segments than typical URLs, length and the number of path segments were added as features. Since the classes were unbalanced, we over-sampled from the minority class using the synthetic minority over-sampling technique (SMOTE) (Chawla et al., 2002). Similarly, for the document model, we used tf-idf features after tokenizing the document using a regex tokenizer and removing stop words. The combined model was a combination of the URL and document features.

To train the RoBERTa model on the privacy policy classification task, we used the

sequence classification head of the pretrained language model from HuggingFace (Wolf et al., 2019). We used the pretrained RoBERTa tokenizer to tokenize text extracted from the documents. Since Roberta accepts a maximum of 512 tokens as input, only the first 512 tokens of text from the documents were used for training while the rest was discarded. As shown in the analysis section, the average length of a privacy policy in terms of the number of words is 1,871. Thus 512 tokens would take into account about a fourth of an average privacy policy.

The 1,600 labelled documents were randomly divided into 960 documents for training, 240 documents for validation and 400 documents for testing. Using 5-fold cross-validation, we tuned the hyperparameters for the models separately with the validation set and then used the held-out test set to report the test results. Due to its size, it was possible for the held out test set to have a biased sample. Thus we repeated the sampling and training processes with a 5-fold cross-validation approach. Table 3.1 shows performance of the models after the results from test sets were averaged. Since the transformer based model had the best results, we ran it on all the the candidate privacy policies. Out of 2.1 million English candidate privacy polices, 1.54 million were classified as privacy policies and the rest were discarded.

Model	Precision	Recall	F1
DBSCAN	0.81	0.83	0.82
URL Based	0.88	0.89	0.88
Document Based	0.93	0.95	0.94
Combined	0.94	0.97	0.95
RoBERTa	0.97	0.98	0.97

Table 3.1: Document classification

3.4.4 URL Cross Verification

Legal jurisdictions around the world require organisations to make their privacy policies readily available to their users. As a result, most organisations include a link to their privacy policy in the footer of their website landing page. In order to focus PrivaSeer Corpus on privacy policies that users are intended to read, we cross-verified the URLs of the privacy policies in our corpus with those that we obtained by crawling the homepages (landing page) of these domains. Between the 8th and 10th November 2019, we crawled the landing pages and pages one hop from the landing pages for all the domains of the URLs in our corpus. We then gathered the URLs satisfying our selection criteria

and cross-verified them with the URLs in our existing corpus. After cross-verifying the URLs, we were left with a set of 1.1 million web pages.

3.4.5 Duplicate and Near-Duplicate Detection

Examination of the corpus revealed that it contained many duplicate and near-duplicate documents. We removed exact duplicates by hashing all the raw documents and discarding multiple copies of exact hashes. Through manual inspection, we found that a number of privacy policies from different domains had very similar wording, differing only by the organisation or website name. We reason that this similarity could be due to the use of privacy policy templates or generators. We also found abundant examples of near-duplicate privacy policies on the same website. We reason that this similarity could be due to the presence of archived versions of privacy policies on the website. Since we aimed to collect a comprehensive corpus of contemporary policies, we only removed similar policies (near-duplicates) from same domain domains.

To remove near-duplicates from within the same domain we used *Simhashing* (Charikar, 2002). Simhashing is a hashing technique in which similar inputs produce similar hashes. After creating shingles (Broder et al., 1997) of size three, we created 64 bit document Simhashes and measured document similarity by calculating the *Hamming distance* (Manku et al., 2007) between document Simhashes of privacy policies within the same domain. We then obtained a list of all pairs of similar documents based on a distance threshold (measured based on the number of differing bits) that was determined after manual examination of a number of pairs of privacy policies. We then filtered the duplicates based on a greedy approach retaining policies that were longer in length. The remaining documents comprised the corpus.

3.5 Corpus Analysis

The PrivaSeer Corpus consists of 1,005,380 privacy policies from 995,475 different web domains. Privacy policies in this corpus have a mean word length of about 1,871 words and range between a minimum of 143 words and a maximum of 16,980 words. Figure 3.2 shows a histogram of lengths of the privacy policies in number of words.

The corpus contains policies from over 800 different top level domains (TLDs). As shown in Figure 3.3 *.com*, *.org*, and *.net* make up a major share of the corpus covering 63%, 5% and 3% respectively. Country-level domains like *.uk*, *.au*, *.ca* and *.du* show the

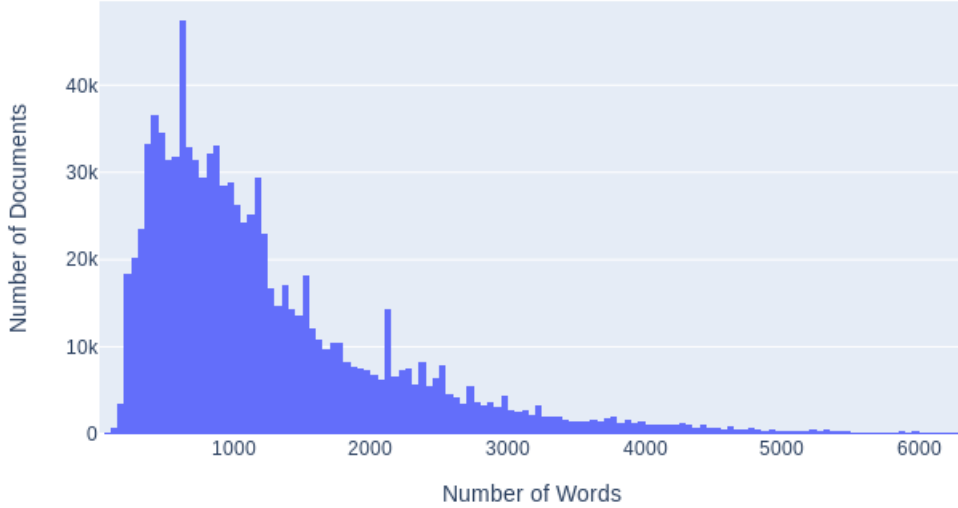


Figure 3.2: Document length distribution

geographic variety of the sources of the corpus covering 12%, 4%, and 2% respectively. The distribution of popular TLDs (.com, .org, .net) roughly matches internet TLD trends suggesting that the corpus contains a random sample of internet web domains. Moreover, CommonCrawl release statistics estimating the representativeness of monthly crawls which support the claim that monthly crawl archives and in turn the PrivaSeer Corpus are a representative sample of the web.

In addition to monthly crawl dumps, Common Crawl releases web graphs with PageRanks of the domains in a crawl. The PageRank values were calculated from the web graph using the Gauss-Seidel algorithm (Arasu et al., 2002). PageRank values can be used as a substitute for popularity where higher values suggest more popular domains.

Figure 3.4 show the distribution of the PageRank values of the domains in the corpus.

3.5.1 Readability

Readability of a text can be defined as *the ease of understanding or comprehension due to the style of writing* (Klare et al., 1963). Along with length, readability plays a role in internet users’ decisions to either read or ignore a privacy policy (Ermakova et al., 2015). While prior studies on readability have shown that privacy policies are difficult to understand for the average internet user, they were conducted using small samples of policies and therefore may not be representative of the larger internet (Fabian et al., 2017). While there are a variety of readability metrics, we calculated the readability

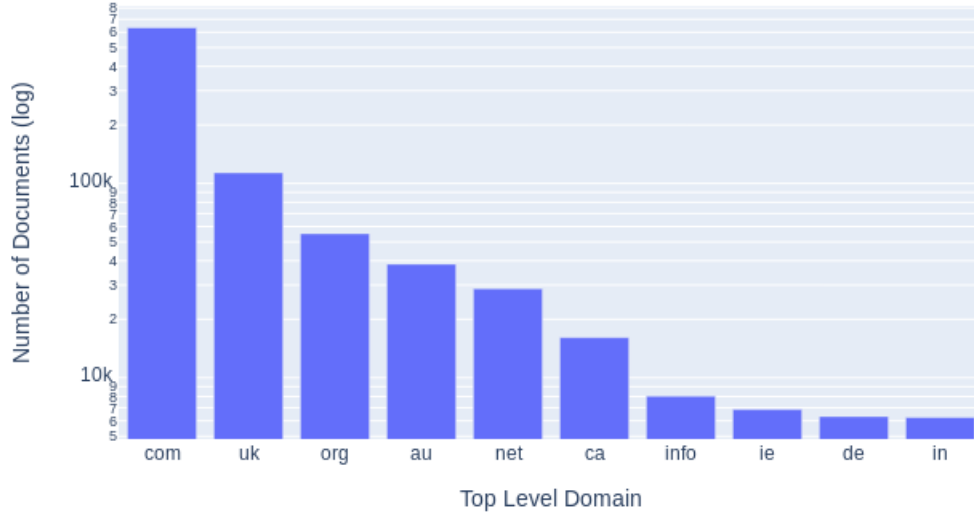


Figure 3.3: Domain distribution



Figure 3.4: Distribution of Policy PageRanks

of the policies in the corpus using the Flesh-Kincaid Grade Level (FKG) metric for comparison with prior literature and since it is the the most widely used metric. The FKG metric presents the readability score as a U.S. grade level. We obtained a mean FKG score of 14.87 and a standard deviation of 4.8. This score can be interpreted as an average of 14.87 years of education in the U.S. (roughly two years of college education) is required to understand a privacy policy. In contrast, Fabian et al. (2017) found that the mean FKG score is 13.6 when they conducted an analysis of readability of privacy policies using 50k documents.

Table 3.2 shows the distribution of the scores. As indicated by the FRES score,

the readability of the privacy policies range from very easy to read (scores between 80 - 100) to very difficult to read (scores between 0 - 20), with the mean score of 40 suggesting that on average privacy policies are moderately difficult to comprehend. The FKG scores suggest that college education is required to understand the average privacy policy. This is corroborated by the SMOG scores which suggest that around 15 years of formal education (e.g., through 2-3 years of college in US education) is required. The most optimistic scores come from CLI which suggests that the average privacy policy is at the reading level of a first-year student in a US college. These results are consistent with prior research and suggest that privacy policies are difficult for an average internet user to read and are at the college reading level.

Metric	Min.	Max.	Mean	Sd.
Words	24	71937	1410.88	1366.46
FRES	-163.41	117.16	40.32	14.20
FKG	-1.9	83.4	14.42	4.36
SMOG	3.1	31.0	15.51	2.27
CLI	2.29	25.83	12.75	1.44

Table 3.2: Average readability scores (Min. = Minimum, Max.= Maximum, Sd. = Standard Deviation)

3.5.2 Topic Modelling

Topic modelling is an unsupervised machine learning method that extracts the most probable distribution of words into topics through an iterative process (Wallach, 2006). We used topic modelling to explore the distribution of themes of text in our corpus. Topic modelling using a large corpus such as PrivaSeer helps investigate the themes present in privacy policies at web scale and also enables the comparison of themes that occur in the rapidly evolving online privacy landscape. We used Latent Dirichlet Allocation (LDA), as our approach to topic modelling (Blei et al., 2003). Since LDA works well when each input document deals with a single topic, we divided each privacy policy into its constituent paragraphs (Sarne et al., 2019), tokenized the paragraphs using a regex character matching tokenizer and *lemmatized* the individual words using NLTK’s WordNet lemmatizer. We experimented with topics sizes of 7, 8, 9, 10, 11, 13 and 15. We manually evaluated the topic clusters by inspecting the words that most represented the topics. We noted that the cohesiveness of the topics decreased as the number of topics increased. We chose a topic size of 9, since larger topic sizes produced markedly less coherent topics.

Interpreted Topic	Vocabulary
First Party Collection/Use	information, service, email, provide, address, use, personal, collect, user, product, name, contact, number
Third Party Collection	google, com, party, third, advertising, ad, website, http, analytics, data, www, user, service, may, network, social, opt, facebook
Data Security and Contact	information, us, personal, security, contact, access, please, secure, data, request, protect, question
Policy Changes	privacy, policy, site, website, use, information, change, service, may, time, term, practice, link
European Audiences	data, personal, processing, right, purpose, subject, consent, protection, legal, interest, process, request, controller, gdpr, legitimate
Cookies and Tracking	cookie, website, use, site, information, browser, may, web, data, user, page, service, visit, computer, used, party, ip
Disclosure of Information	information, service, may, party, third, use, personal, provide, product, purpose, us, company, business, disclose, customer

Table 3.3: Vocabulary and interpreted topics for LDA based topic modelling

Prior work on analysing the themes in privacy policies was undertaken by Wilson et al. (2016). The authors introduced a labeling scheme of privacy practices through the OPP-115 Corpus based upon input from legal experts. They followed a bottom-up approach and identified different categories from analysis of data practices in privacy policies. The categories identified in the OPP-115 Corpus can be found in Table 3.4. In contrast, we followed a top-down approach and applied topic modelling to the corpus in order to extract common themes for paragraphs.

For each topic, we identified a corresponding entry from the OPP-115 annotation scheme (Wilson et al., 2016), which was created by legal experts to label the contents of privacy policies. While Wilson et al. (2016) followed a bottom-up approach and identified different categories from analysis of data practices in privacy policies, we followed a top-down approach and applied topic modelling to the corpus in order to extract common themes for paragraphs. The categories identified in the OPP-115 Corpus can be found in Table 3.4. The top vocabulary for each topic and its manually interpreted category name are shown in Table 3.3. The terms listed in the vocabulary column are in decreasing order of probability, cut off at a threshold. While most of these topics match the coarse-grained level of the OPP-115 scheme, we found that a few topics matched finer details of the scheme.

We found that two LDA topics contained vocabulary corresponding with the OPP-

115 category *First Party Collection/Use*, one dealing with purpose and information type collected and the other dealing with collection method. Two LDA topics corresponded with the OPP-115 category *Third Party Sharing and Collection*, one detailing the action of collection, and one explaining its purpose and effects (advertising and analytics). One of the LDA topics exclusively comprised of vocabulary related to cookies which could be related to both first party or third party data collection techniques. The OPP-115 categories *Privacy Contact Information*, *Data Security* and *Policy Change* appeared as separate topics while a topic corresponding to the OPP-115 category *International and Specific Audiences* appeared to be primarily related to European audiences and GDPR.

The topics *First Party Collection/Use*, and *Policy Changes* match categories in the OPP-115 Corpus. The topics *Third Party Collection*, *Cookies and Tracking*, and *Disclosure of Information* are all attributes of data practices for the ‘Third Party Sharing/-Collection’ category in the OPP-115 Corpus, suggesting that privacy policies separately mention the above three topics instead of grouping them together. The topic *European Audiences*, appears as a attribute of in the ‘International and Specific Audiences’ category in the OPP-115 Corpus. The topic *Data Security and Contact* appears to be a combination of the OPP-115 Corpus categories ‘Data Security’ and ‘Other’.

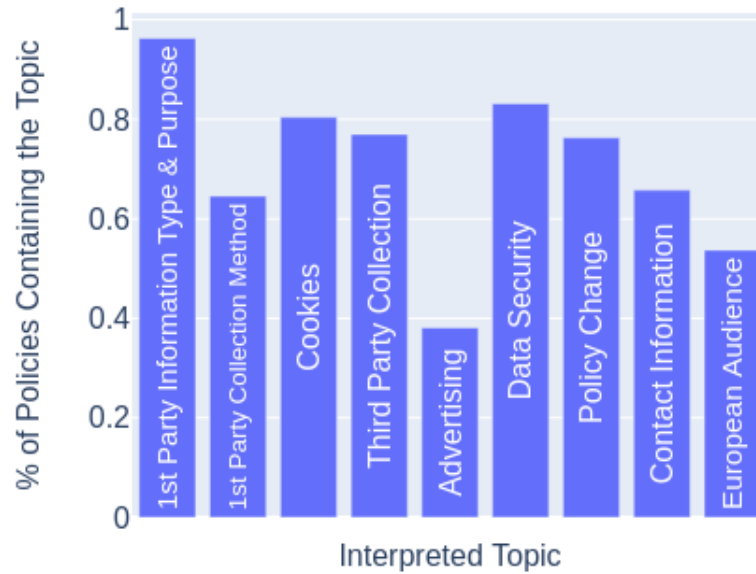


Figure 3.5: Topic distribution

It is likely that the divergence between OPP-115 categories and LDA topics comes from a difference in approaches: the OPP-115 categories represent themes that privacy experts expected to find in privacy policies, which diverge from the actual distribution

of themes in this text genre. Figure 3.5 shows the percentage of privacy policies in the corpus that contain each topic. From the figure we see that information regarding the type and purpose of data collected by first and third party sources are the most common topics. About 77% of policies contain language regarding third parties. This is consistent with prior research on third party data collection (Libert, 2018). In contrast, language regarding advertising and analytics appears in only 38% of policies in the corpus. Topics corresponding to data security, policy change and contact information also occur in a majority of privacy policies. Language corresponding to the GDPR and European audiences appears in 55% of policies. A study of the distribution of privacy policy topics on the web is important since they inform us about real-world trends and the need for resource allocation to enforce of privacy regulations.

Figure 3.6 shows how the number of topics in privacy policies vary with respect to the PageRank value. The whiskers in the plot represent the 95% confidence interval of the means of the number of topics in the privacy policies in each PageRank value bin. The PageRank values were binned with a constant value of 0.25 such that each bin had at least 1k privacy policies. The plot suggests that more popular domains (as given by PageRank value) tend to address a greater number of topics in their privacy policies. This behaviour is consistent with manual inspection and is likely due to a larger and more diverse user base as well as the greater levels of regulatory scrutiny that accompany it in the case of more popular domains. For example, popular organisations tend to be multinational thereby requiring to address privacy laws from multiple jurisdictions such as GDPR from the European Union and CCPA from the United States. We found a similar pattern between privacy policy length and PageRank value thereby further supporting our claim that the more popular domain privacy policies tend to address a greater number of topics. In addition we found that readability and PageRank follow a similar pattern where privacy policies of more popular domains (as given by PageRank values) tend to be slightly more difficult to read.

Figure 3.7 shows how the readability using FKG varies with respect to the domain PageRank values. The approach for the whiskers and bins in the plot is similar to that of Figure 3.6. The whiskers in the plot represent the 95% confidence interval of the means of the readability of privacy policies in each PageRank bin. The PageRank values were binned with a constant value of 0.25 such that each bin had at least 1000 privacy policies. The plot suggests that on average, privacy policies on more popular domains (as given by PageRank values) tend to be slightly more difficult to read. As seen in the case of topic modelling this might be explained by more popular domains having higher

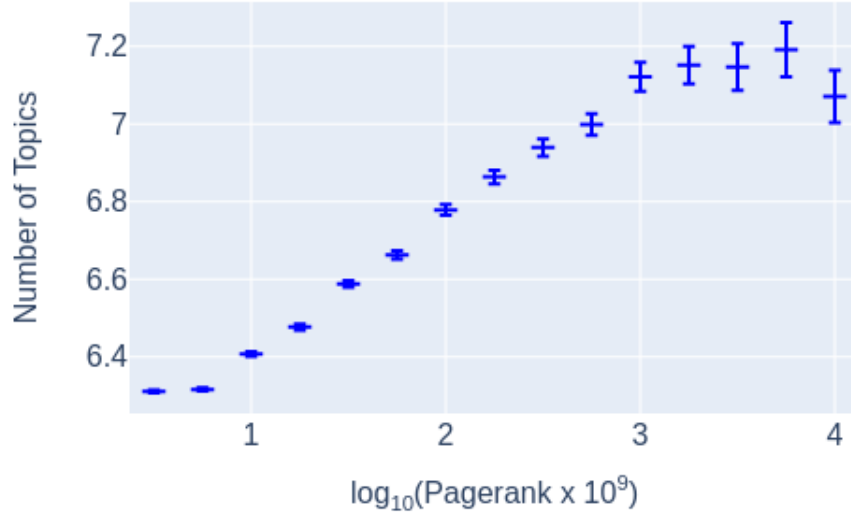


Figure 3.6: Relationship between number of topics and privacy policy domain PageRank value

coverage of privacy policy topics.

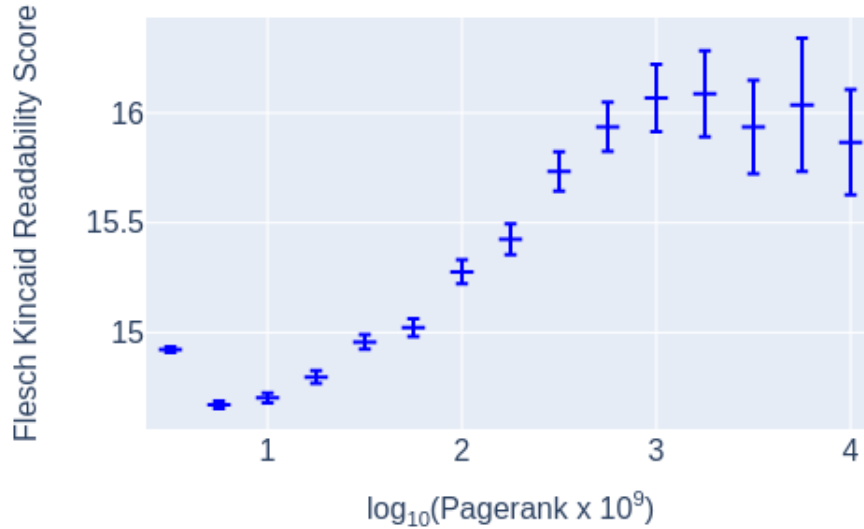


Figure 3.7: Relationship between privacy policy readability and domain PageRank value

3.5.3 Similarity of Web Privacy Policies

While manually analysing privacy policies in the corpus, we noted that various privacy policies from different domains/organisations shared common passages of text, differing only by the organisation name. In order to further investigate, we used the *Jaccard*

Similarity Index to determine textual similarity between privacy policies. We uniformly randomly sampled 1,000 documents from the corpus and created shingles of window size 3 for each documents. We then calculated the Jaccard Similarity Index between all the document pairs in our sample and obtained a total of 1,000,000 document pair similarities. To interpret the degree of similarity between a document pair based on its Jaccard Similarity Index, we split all the documents into their component sentences. We then compared the pair’s Jaccard Similarity Index to the ratio of the number of identical sentences between the pair of documents to the total number of sentences in the smaller of the two documents. We found that the two factors followed a linear relationship with a correlation of 0.78 and p-value less than 0.001. For example, given that two documents are of the same size, if the Jaccard Similarity Index between a pair of documents is 0.5, we found that that on average about 40% of the sentences between them are identical.

We then calculated the similarity of 1 million document pairs using the Jaccard Similarity Index. We found that while each document in the sample is dissimilar to most of the other documents in the sample, with a Jaccard Similarity Index less than 0.1, many documents are very similar to a few of other documents, with a Jaccard Similarity Index greater than 0.5. Additionally, we found that a large number of small clusters with 2 to 10 documents in a cluster with most other documents relatively far apart. High Jaccard Similarity Indexes greater than 0.7 are likely due to identical privacy policies only differing by organisation name. Similarity between groups of privacy policies is likely to be the result of privacy policy generators or privacy policy templates. These results suggest that a significant number of privacy policies on the web might be created from fixed templates, suggesting new avenues for information extraction from privacy policies.

3.6 PrivBERT

In order to address the requirement of a language model for the privacy domain, we created PrivBERT. BERT is a contextualized word representation model that is pre-trained using bidirectional transformers (Devlin et al., 2019). It was pretrained on the masked language modelling and the next sentence prediction tasks and has been shown to achieve state of the art results in many NLP tasks. RoBERTa improved upon the results achieved by BERT by making improvements to the training technique (Y. Liu et al., 2019). We pretrain *PrivBERT* starting with the pretrained RoBERTa_{BASE} model (12 layers, 768 hidden size, 12 attention heads, 110M parameters). RoBERTa was trained on

corpora of books, news articles, Wikipedia and social media comments and works well as a general purpose language model. Privacy policies written in legalese differ significantly in language when compared to the corpora used to train BERT and its variants, thereby prompting the need for a separate pretrained language model. Prior literature has shown that in-domain language models such as SciBERT (Beltagy et al., 2019) and BioBERT (Lee et al., 2020) perform significantly better on tasks in their respective domains.

Label	Polisis	RoBERTa			PrivBERT			Support
	F1	P	R	F1	P	R	F1	
First Party Collection and Use	0.82	0.92	0.89	0.91	0.95	0.88	0.92	250
Third Party Sharing and Collection	0.82	0.86	0.91	0.88	0.88	0.93	0.91	203
User Choice/Control	0.72	0.83	0.78	0.80	0.87	0.79	0.83	77
Privacy Contact information	0.84	0.84	0.76	0.79	0.82	0.80	0.81	42
Introductory/Generic	0.73	0.88	0.67	0.76	0.79	0.76	0.77	78
Practice Not Covered	0.13	0.50	0.32	0.39	0.65	0.44	0.52	25
Data Security	0.75	0.91	0.75	0.82	0.94	0.80	0.86	40
User Access, Edit and Deletion	0.70	0.72	0.88	0.79	0.80	0.88	0.84	24
Policy Change	0.88	0.79	0.90	0.84	0.87	0.95	0.91	21
Do Not Track	1.0	1.0	1.0	1.0	1.0	1.0	1.0	3
International and Specific Audiences	0.82	0.83	0.78	0.81	0.89	0.84	0.86	56
Data Retention	0.40	0.80	0.57	0.67	0.83	0.71	0.77	14
Macro Averages	0.71	0.82	0.77	0.79	0.86	0.82	0.83	833
Micro Averages	0.78	0.86	0.82	0.84	0.88	0.85	0.87	833

Table 3.4: Test performance comparison of three models on the data practice classification task (P:Precision, R: Recall)

We use the byte pair encoding tokenization technique utilized in RoBERTa and retain its cased vocabulary. We did not create a new vocabulary since the two vocabularies are not significantly different and any out-of-vocabulary words can be represented and tuned for the privacy domain using the byte pair encoding vocabulary of RoBERTa. We preprocessed the privacy policy documents to create sequences of a maximum length of 512 tokens. Inputs significantly shorter than the maximum length occasionally occurred since we did not create sequences that crossed document boundaries. We trained PrivBERT using dynamic masked language modelling (Y. Liu et al., 2019) for 50k steps with a batch size of 512 using the gradient accumulation technique on two NVIDIA Titan RTX for 8 days with a peak learning rate of 8e-5. Other hyperparameters were set similar to RoBERTa.

3.6.1 Finetuning PrivBERT

We evaluated the performance of PrivBERT on two tasks: (i) Data practice classification (ii) Answer sentences selection.

For the data practice classification task, we leveraged the OPP-115 Corpus introduced by Wilson et al. (2016). The OPP-115 Corpus contains manual annotations of 23K fine-grained data practices on 115 privacy policies annotated by legal experts. To the best of our knowledge, this is the most detailed and widely used dataset of annotated privacy policies in the research community. The OPP-115 Corpus contains paragraph-sized segments annotated according to one or more of the twelve coarse-grained categories of data practices. We fine-tuned PrivBERT on the OPP-115 Corpus to predict the coarse-grained categories of data practices. We divided the corpus in the ratio 3:1:1 for training, validation and testing respectively. Since each segment in the corpus could belong to more than one category and there are twelve categories in total, we treated the problem as a multi-class, multi-label classification problem. After manually tuning hyperparameters, we trained the model with a dropout of 0.15 and a learning rate of $2.5e-5$.

Table 3.4 shows the results for the data practice classification task comparing the performance between RoBERTa, PrivBERT and Polisis (Harkous et al., 2018), a CNN based classification model. We report reproduced results for Polisis since the original paper takes into account both the presence and absence of a label while calculating the score for each label Nejad et al. (2020). Due to the unbalanced nature of the dataset, we report the macro-average and micro-average scores. PrivBERT achieves state of the art results improving not only on the macro-average F1 score of RoBERTa by about 4% but also improving on the F1 score for every category in the task.

For the question answering task, we leveraged the PrivacyQA corpus (Ravichander et al., 2019). PrivacyQA consists of 1,750 questions about the contents of privacy policies from 35 privacy documents. While crowdworkers were asked to come up with privacy related questions based on public information about an application from the Google Play Store, legal experts were recruited to identify relevant evidence within respective privacy policies that answered the question asked by the crowdworkers. The goal of the question answering task is to identify a set sentences in the privacy policy that has information relevant to the question. Ravichander et al. (2019) divided the corpus into 1,350 questions for training and validation and 400 questions for testing where each question in the test set is annotated by at least three experts. We fine-tuned PrivBERT on the training set as a binary classification task on each question-answer sentence pair to identify if the sentence is evidence for the question or not. We trained the model with a dropout of 0.2 and a learning rate of $3e-6$ with the positive and negative classes weighted in the ratio 8:1 during training. We used sentence level F1 as the evaluation metric as

described by Ravichander et al. (2019), where precision and recall are calculated by measuring the overlap between the predicted sentences and gold standard sentences.

Model	Precision	Recall	F1
BERT	0.442	0.348	0.39
RoBERTa	0.31	0.51	0.41
PrivBERT	0.483	0.424	0.452

Table 3.5: Performance comparison on the answer sentence selection task

Table 3.5 shows the results for the answer sentence selection task comparing the performance between BERT and PrivBERT. Results from BERT are as reported by Ravichander et al. (2019). PrivBERT achieves state of the art results improving on the results of BERT by about 6%. PrivBERT therefore has been shown to achieve state of the art results in two significantly disparate tasks in the privacy domain suggesting that it can be used to improve the performance on various real-world tasks and application in the privacy domain.

3.7 Conclusion

We created the PrivaSeer Corpus which is the first large scale corpus of contemporary website privacy policies and consists of just over 3 million documents. We designed a novel pipeline to build the corpus, which included web crawling, language detection, document classification, duplicate removal, document cross verification, content extraction, and near duplicate removal.

Topic modelling showed the distribution of themes of privacy practices in policies, corresponding to the expectations of legal experts in some ways, but differing in others. The positive relationship between PageRank of a domain and the number of topics covered in its policy indicates that more popular domains have a slightly greater coverage of these topics. We hypothesize that this is because more popular domains tend to have a larger and more diverse user base prompting the privacy policies to address laws from various jurisdictions.

Prior research on the readability based on small corpora of privacy policies had found that they were generally hard to understand for the average internet user. Our large scale analysis using the Flesch-Kincaid readability metric was consistent with prior findings. We found that on average about 14.87 years or roughly about two years of U.S. college education was required to understand a privacy policy. In addition, we found that

privacy policies of more popular websites are slightly harder to read. We hypothesize that the reasoning behind this finding follows a similar finding in the topic modelling task.

We pretrained PrivBERT a language model for the privacy domain based on RoBERTa. We evaluated PrivBERT on the data practice classification and the question answering tasks and achieved state of the art results.

We created PrivaSeer, the first privacy policy search engine. Documents can be ranked by query based document relevance scores, PageRank values, and document probabilities. They also can be filtered based on sector of commerce, policy vagueness, policy readability, tracking technology mentioned, regulatory bodies mentioned, and regulations/cross-border agreements mentioned in the policy text.

On average about 30% of the sentences in a privacy policy were found to have at least one vague word in them. This suggests that vagueness in privacy policy documents is a pervasive problem. We used regex text matching to extract details about tracking technology, regulatory bodies, and regulations/cross-border agreements and found non instances of false positives. We believe this is because privacy policies only record elements of privacy that they use/comply while rarely mentioning other elements/alternatives that exist.

An exploratory evaluation of PrivaSeer based on PageRank, query based relevance, and our custom ranking technique found that the custom ranking technique outperformed the others in all categories. We found that our custom technique had higher variability in returned results and was able to overcome limitations caused by the presence of false positive privacy policies in the results.

We believe that the PrivaSeer Corpus will help advance research techniques to automate the extraction of salient details from privacy policies. PrivBERT will help improve results on various tasks in the privacy domain and help build stable and reliable privacy preserving technology. The PrivaSeer search engine is a necessary tool that is the first of its kind and is helpful to several distinct groups with goals in furthering user privacy. This should benefit internet users, regulators, and researchers in many ways.

Chapter 4 —

Investigation of Policy Availability

4.1 Background

Across the world, privacy policies tend to be the primary and often the only source of information regarding what happens to users' personal information online. The GDPR (Articles 12-14), explicitly requires transparent disclosure of information, accurately reflecting data practices. The CPRA further formalizes this requirement, making it mandatory for organizations that exceed a \$25M annual revenue or process information of more than 50k users to post a privacy policy and update it at least once a year (Assembly, 2020). However, policy regulators are often overwhelmed by the number of privacy policies on the web. Policy regulators even in privacy-friendly jurisdictions, such as the EU, rely on user complaints to investigate privacy practices (Supervisor, 2018) while others (such as the United States) rely on organizations to self-certify their compliance¹ and only investigate when a privacy policy is at odds with real-world privacy practices. It is therefore essential that all organizations make their privacy policies easily accessible to users so that they can make an informed decision about their privacy online. To this end, we investigate the lack of availability of policies by studying the frequency of various failure modes in privacy policy availability, due to issues such as broken links, empty content, placeholder text and language issues. Our analysis of trends in failure modes of privacy policies will throw light on the issues in online privacy landscape thereby motivating further study. Lack of availability of privacy policies is especially worrying since it signifies a breakdown of *notice and choice* and therefore warrants careful investigation. Next, we undertake the task of estimating the lower bound on the total number of privacy policies on the web using the capture-recapture (Lincoln-Petersen

¹<https://www.privacyshield.gov/Program-Overview>

estimator) technique (Begon et al., 1979) in order to study the trends in distribution of policies on the internet. We estimate the distribution of website privacy policies in terms of sectors of commerce and the most popular top level domains (TLDs).

To undertake this study, we begin with a collection of company websites and their corresponding sectors of commerce from LinkedIn. We then crawl the websites’ landing pages and their corresponding privacy policies (if one exists) and take a note of the various error modes we encounter. Additionally, we manually sample and investigate 500 candidate privacy policy pages. We find users are likely to find a privacy policy in less than 34% of websites due a number of failure modes such as absence of policy, dead links, empty content, placeholder text and language discrepancies. This brings into question the efficacy of the notice and choice as a framework for user privacy online. Next, we estimate the total number of privacy policies on the web and their distribution based on sectors of commerce and various top level domains thereby showing light on the trends of policies online. We find the lower bound on the number of English language privacy policies on the web is around 3 million based on the capture-recapture estimation technique.

4.2 Related Work

4.2.1 Privacy Policy Issues

Previous investigations on the insufficiency of privacy policies have mainly concentrated on their complex language and the reasons behind users’ reluctance to read them. Nonetheless, scrutiny of privacy policies has revealed a concerning silence on vital consumer-related practices, such as the collection and utilization of sensitive information, as well as the integration of tracking data with personally identifiable information. Many of these policies have been found to lack full compliance with self-regulatory guidelines (Hoke et al., 2015; L. F. Cranor et al., 2014). Moreover, analysis of policies from 16 of the globe’s largest internet and telecommunications companies has indicated that vague or unclear language impedes users’ comprehension of company practices, hampering informed decision-making on products and services (P. C. Kumar, 2016). The implementation of the General Data Protection Regulation (GDPR) in May 2018 has played a pivotal role in ensuring online privacy rights for users (Kretschmer et al., 2021). Studies have further investigated the factors that correlate with GDPR compliance practices within organizations by examining the corresponding privacy policies (Aberkane et

al., 2022).

In this chapter, we investigate a more fundamental question about trends in the ability of access to these documents. Another similar line of research involved identifying mismatches between user expectations and privacy practices stated in privacy policy documents (Rao et al., 2016). This work described the potential of highlighting unexpected practices in websites thereby helping users to make better privacy decisions. Similarly, Sunyaev et al. assessed the availability, scope, and transparency of privacy policies of mobile health apps on iOS and Android and showed that privacy policies have poor transparency and availability rates. They found that of the 600 most commonly used apps, only 183 (30.5%) had privacy policies (Sunyaev et al., 2014).

4.2.2 Document Estimation

The capture-recapture technique, also known as the Lincoln-Petersen estimator or mark-recapture, is a solution to the problem of estimating the size of a population when it is not feasible to count every individual item within it. This technique has traditionally been utilized in ecology to estimate animal populations (Begon et al., 1979) and in epidemiology to estimate disease prevalence (Brenner, 1995). For example, the capture-recapture technique was employed by Bohning et al. to estimate the number of COVID-19 infections (Böhning et al., 2020). In the realm of information security, Weaver et al. utilized this technique to gauge the extent of phishing activity on the internet (Weaver & Collins, 2007). In information retrieval, researchers have applied this technique to estimate the size of the world wide web (Lu & Li, 2010) and the coverage of major search engines (Lawrence & Giles, 1998; Dobra & Fienberg, 2004). Khabsa and Giles (Khabsa & Giles, 2014) estimated the number of scholarly documents available on the web and the size of Google Scholar using this method. In this paper, we estimate the number of English language privacy policies on the web in order to inform trends in the online privacy landscape.

4.3 Alternate Privacy Policy Corpus Creation

In order to create an alternate privacy policy corpus, we begin with a list of company domains from the Free Company Dataset² provided by People Data Labs. The Free Company Dataset is a collection of over 7 million global companies containing fields such

²<https://docs.peopledatalabs.com/docs/free-company-dataset>

as name, domain, year founded, industry, size range, locality, country, LinkedIn URL, current employee estimate, and total employee estimate. We then follow the pipeline described in chapter 3. We extracted a list of company domain URLs from the dataset and obtained about 5.5 million unique URLs. We then used Scrapy³ to crawl the URLs. About 1.5 million domain URLs returned an error while 4 million were successfully crawled. Privacy laws around the world have created an industry standard to include a link to the privacy policy in the footer of the website landing page. Additionally, we found that privacy policy URLs tend to have the word 'privacy' or the words 'data' and 'protection' in them. Thus we extracted candidate privacy policy URLs from website domain pages by searching for HTML *href* attributes containing the words *privacy* or the words *data* and *protection*. We refer to this as our **URL selection criteria**. The majority of the successful domain URLs did not produce a successful potential privacy policy URL since they either did not have a privacy policy, or none of the the hyperlinks in the page satisfied our URL selection criteria. We crawled a total of about 950k web pages that satisfied our URL selection criteria. We refer to these web pages as **candidate** privacy policy documents. Next, we put the candidates policies through LangID, a language identification tool (Lui & Baldwin, 2012b). We used LangID due to its high accuracy over a range of languages and domains. LangID accepts a segment of text and returns the identified language and is capable of detecting 97 languages. We retained the 790k candidates were identified to in English for further analysis and discarded the rest.

In order to separate privacy policies from other types of documents that only satisfied our URL selection criteria, we used a machine learning approach. We uniformly randomly sampled 1,600 English documents and labelled them to be either a privacy policy or not. We then trained a random forest model with features extracted from the policy document and URL. We separately tokenized the words in the policy document and policy URL and removed stop words. We then calculated the term frequency-inverse document frequency (tf-idf) for each policy. We divided the labelled documents into train, validation and test sets in the ratio 3:1:1. After tuning the hyperparameters using the validation set and a 5-fold cross-validation, the precision, recall and F1 scores were found to be 0.96, 0.97 and 0.96 respectively. We ran the trained random forest classifier on the English language candidate privacy policies and found that 625k documents were classified as privacy policies while the other 165k were not privacy policies.

We removed duplicates and near-duplicates from the 625k documents classified as

³<https://scrapy.org/>

privacy policies. Duplicates were removed by hashing the downloaded privacy policy files and discarding privacy policies that generated a duplicate hash. To remove the near-duplicates, we first grouped privacy policies belonging to the same domain. We then applied the simhash (Charikar, 2002) algorithm on shingles of size three for each privacy policy. Hamming distance between each pair of privacy policies within the same domain was calculated using the simhashes and near-duplicates were filtered out based on a Hamming distance threshold (Manku et al., 2007). We obtained 612,046 unique English language privacy policies from the Free Company Dataset. We denote this set of privacy policies as the CoLIPPs (Companies on LinkedIn Privacy Policies) Corpus.

4.3.1 Evaluation of URL Selection Criteria

In order to estimate the accuracy of our URL section criteria, we manually examined a random sample of 500 company URLs, from the 4 million successfully crawled domain URLs. Table 4.1 lists our observations from this sample. We found that 330 domains did not contain a hyperlink to a privacy policy, while 170 domains listed one on their landing page. Among the 170 policy URLs found, 106 contained the chosen keywords (*privacy* or *data* and *protection*) as is, while 42 had the chosen keywords (as well as the privacy policy itself) in other languages. Additionally, 22 websites displayed their policies with JavaScript in HTML elements such as a dialog box. Out of these, 17 documents were in English.

Our work focuses on collecting English language privacy policies. Therefore, while estimating the accuracy of our URL section criteria, we exclude the websites that were in languages other than English. Thus, given that 106 English language websites had the chosen keywords in their privacy policy URL, out of a total of 123 English language websites, we estimate that our URL selection criteria captured the privacy policies of 86.17 ± 7.38 % of English language websites. We also found that only 34 ± 4.15 % of websites in the sample had a privacy policy hyperlink on the landing page irrespective of language.

4.4 Policy Availability Analysis

In this section, we examine three distinct failure modes of privacy policy availability, namely dead/broken links, language discrepancies, and non-privacy policy documents.

Observation		Number
Websites without a privacy policy hyperlink		330
Websites with a privacy policy hyperlink	Chosen keywords in privacy policy URL	106
	Translated versions of chosen keywords in privacy policy URL	42
	Privacy policy displayed using other HTML elements	English: 17 Others: 5
Total		500

Table 4.1: Observations on a random sample of 500 company URLs from successfully crawled domain URLs

4.4.1 Dead Links to Privacy Policies

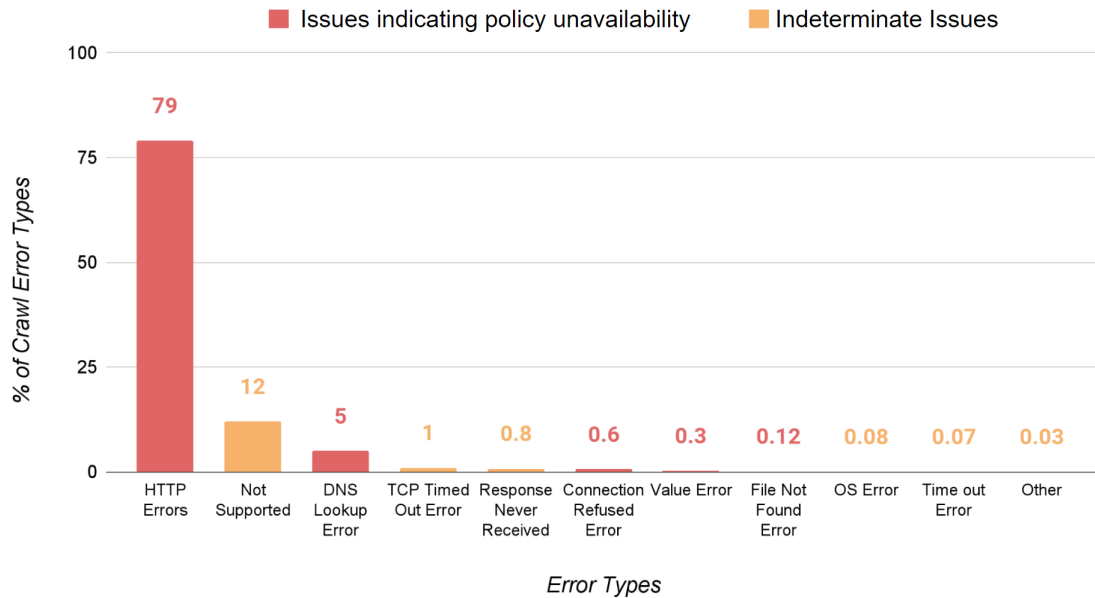


Figure 4.1: Crawling Errors

Figure 4.1 shows error types that had at least 10 instances when crawling the URLs that satisfied our selection criteria. The figure shows the percentage of each error type that were issued by the server in response to a request made by the crawler. The error Types *HTTP Error*, *DNS (Domain Name Server) Lookup Error*, *Connection Refused Error*, *Value Error*, and *File Not Found Error* indicate unavailability of privacy policies, since these error types indicate issues with the availability of a web page. The error

types *Not Supported* (crawler not being able to perform an action on HTML elements such as JavaScript pop-ups), *TCP Timed Out*, *Response Never Received*, *OS Error* and *Timeout Error* could be due to any number of issues with crawling and are therefore not indicative of unavailable pages.

13,261 URLs failed with *HTTP Errors*, with a majority returning *404 (Page Not Found)* and *500 (Internal Server Error)*, both indicative of an error in configuring the requested page. 764 URLs failed with *DNS Lookup Errors*; since the URLs were obtained from successfully crawled domains, a DNS error suggests that the URLs were invalid. 95 instances of *Connection Refused Errors* were observed; manual inspection revealed that these errors were either due to URLs referencing *localhost* or due to incorrectly configured ports. 51 URLs had *Value Errors*; manual examination showed that they were comprised of invalid URLs. 17 links returned *File Not Found Errors*, which suggested that they had invalid configurations. Summing all the errors due to the above error types, we found that 14,188 URLs could possibly be due to unavailable web pages.

To investigate the nature of the 14,188 error URLs that were likely due to unavailable privacy policies, we randomly sampled 100 and found that 98 URLs had all the markers of a privacy policy URL i.e. containing the term 'privacy', present in the footer of the landing page. This suggests that a conservative estimate of 13,189 error URLs (95% confidence interval) did not redirect to a valid privacy policy page. We therefore estimate that 13,189 out of all the candidate policies or 1.37% of all privacy policy URLs irrespective of language are dead links. It should be noted that this is a conservative estimate, since we calculate the percentage of dead links with respect to the collection of candidate privacy policy URLs not actual privacy policy URLs.

4.4.2 Natural Language Discrepancies

With many websites being offered in multiple languages, we investigated whether the websites which were offered in multiple languages also offered their privacy policy in those languages. We identified inconsistencies where privacy policy text was unavailable in certain languages. For example, in one particular case the website was displayed in English with options to switch to Spanish and Portuguese. When we attempted to access the privacy policy in the English version, we were directed to the document in Spanish. This behaviour certainly does not help users not literate in Spanish, who would be unable to read the document without translation services.

To estimate such inconsistencies, we randomly sampled 250 domain URLs from the 4 million domains which were successfully crawled. 31 out of the 250 domains were avail-

able in multiple languages, with English always being one of the languages in which the website was offered. 13 domains presented inconsistent user experience by not displaying the privacy policy document in a preferred language set on the landing page. Only 10 domains offered the privacy policy in all the languages in which the rest of the website was offered, while the remaining 8 did not have a privacy policy in any language. Thus, we find that $41.9 \pm 17.35\%$ of the websites that are offered in multiple languages have privacy policy language inconsistencies, and in general we estimate that $5.2 \pm 2.4\%$ of all the websites (where at least one of the languages is English) display this type of language inconsistency with respect to privacy policies.

Additionally, we found that about 2,858 were in Latin, a language that is unsuitable for legal notice and choice in any modern jurisdiction. We informally examined the contents of these documents and found most consist of placeholder texts, such as *Lorem ipsum* (Wikipedia contributors, 2020).

4.4.3 Non-Privacy Policy Documents

20.71% (163,049) of candidate privacy policy documents in English were classified as non-privacy policy documents. Manual inspection of these documents revealed that they included documents with valid privacy policy URLs. To identify the reason for the classification, we examined the content of these documents. We found that a number of valid privacy policy URLs led to web pages that had empty privacy policies, i.e., they had the heading *Privacy Policy* but did not have any content.

To estimate the rate of occurrence of empty privacy policies, we randomly sampled 500 non-privacy policy documents and manually inspected their content. $6.8 \pm 2\%$ (34) of documents in the random sample had empty content on the privacy policy page suggesting that 7,753 (considering the lower bound) documents out of the 163,049 documents classified as not a privacy policy are in reality privacy policy web pages without any content. We estimate that on average, 1.23% of all valid English language privacy policy URLs have empty content.

4.5 Privacy Policy Estimation

To estimate the number of English language privacy policies on the web, we used the Lincoln-Petersen estimator (Pollock et al., 1990), commonly used to estimate wildlife species population in ecological studies. In this technique, a sample of animals from a

TLD	Privaseer Corpus	CoLIPPs Corpus	Overlaps	Estimate	% Share
.com	632,971	387,377	130,168	1,883,707	63.1
.uk	113,515	78,353	27,149	327,608	10.99
.org	55,133	34,832	13,630	140,895	4.73
.au	38,420	30,856	10,329	114,839	3.85
.net	28,718	11,830	3853	88,588	2.97
.ca	16,123	9552	3227	47,724	1.6
.info	8049	576	173	26,799	0.89
.in	6251	3843	928	25,886	0.86
.ie	6884	4288	1509	19,505	0.65
.de	6321	2835	929	19,290	0.64

Table 4.2: Top 10 TLDs with the greatest number of privacy policy documents.

population is captured, marked, and released back into the wild. Then, a second set of animals from the same population is captured and the number of previously marked animals are counted. The total number of animals in the population is then estimated based on the intuition from Bayes rule that the ratio of number of animals captured in the first sample to the total population is equal to the ratio of marked animals recaptured to the number of animals captured in the second sample, given that the two samples are random and independent.

We used privacy policies collected from two separate crawls as our two samples. We used the PrivaSeer Corpus (Srinath, Wilson, & Giles, 2021) as the first sample and the CoLIPPs Corpus that we created as the second sample. The PrivaSeer Corpus is a collection of 1,005,380 English language privacy policies crawled in July 2019 using URLs extracted from CommonCrawl. CommonCrawl releases monthly crawl archives which provide a representative sample of the web⁴. Since CommonCrawl monthly archives are collected independent of the Free Company Dataset, which we used to build the CoLIPPs Corpus, we assume that the PrivaSeer Corpus and the CoLIPPs Corpus represent independent samplings.

To estimate the number of English language privacy policies on the web, we extracted all the URLs from the PrivaSeer Corpus (n_1) and counted them as marked samples. We then extracted all the URLs from the CoLIPPs Corpus (n_2). Next, we counted the number of recaptured (marked in the first sample) URLs i.e. the URLs that are present both in the PrivaSeer Corpus and the CoLIPPs Corpus (m) and estimated the total number of English language privacy policies on the web (N), including a confidence interval, using the following formulae (Pollock et al., 1990).

⁴<https://commoncrawl.github.io/cc-crawl-statistics/>

$$N = \frac{n_1 \cdot n_2}{m} \quad (4.1)$$

$$100(1 - \alpha)\% = \frac{n_1 \cdot n_2}{m} \pm z_{1-\alpha} \sqrt{\frac{n_1 \cdot n_2 \cdot n_{12} \cdot n_{21}}{m^3}} \quad (4.2)$$

where n_{12} is the number of objects captured in the first sample but not the second and n_{21} is the number of objects captured in the second sample but not the first.

The Lincoln-Petersen technique makes three assumptions for estimation:

1. That the population is closed to additions and deletions: This means that between the two samples, no new objects can be added or removed. The Lincoln-Petersen technique, however allows for unmarked additions, and deletions if they occur randomly with respect to marked and unmarked objects.
2. That marks are not lost or altered such that an object marked in the initial sample could not be re-identified if later recaptured.
3. That all objects are equally likely to be captured.

With respect to the assumption (1), although the number of privacy policies on the web is not a closed population, all privacy policies added after the first sample were not previously crawled, therefore we can consider them to be unmarked. We can also assume that deletions occurred randomly between marked and unmarked privacy policies due to the constant flux in the number of websites and therefore privacy policies on the web. Thus, we can assume that assumption (1) is satisfied.

We tailored our marking technique to satisfy assumption (2). We marked the privacy policies based on the domain name found in its website URL, i.e., if the domain name of the privacy policy URL in the PrivaSeer corpus was found in the CoLIPPs Corpus, then it was counted as a recapture even if the URL paths were different. We followed this technique since the two corpora were crawled seven months apart, during which several privacy policy URL paths within a website might have changed. In order to use domain names as markers, we made the additional assumption that all domains have at most one privacy policy, since the vast majority of websites on the internet have only one valid current privacy policy.

With respect to assumption (3), both the selected corpora use a URL selection criteria to filter privacy policy candidates. The URL selection criteria excludes all URLs which do not have the word *privacy* or the words *data* and *protection* in them. We

estimated that this selection criteria excludes a maximum of about 21.2% (from the evaluation of the URL selection criteria in Section 3) of the English language privacy policies on the web. Thus, compared to a random privacy policy online, it is more likely that the objects marked in the first sample are recaptured in the second. We also acknowledge that the sources for both the corpora might have their own biases. For example, CommonCrawl, the source for the PrivaSeer Corpus, create their monthly archives based on seed URLs from various sources and uncrawled domains from previous archives. Thus, the domains crawled might be biased towards the original seed URLs based on the method of randomization. Similarly, the CoLIPPs Corpus was derived from a corpus created by crawling LinkedIn. Thus the domains in the corpus might be biased towards organisation/company domains. Although this bias might partially be offset by the size of the corpora, it is unlikely to be completely removed. This would therefore cause an underestimation of the total number of privacy policies on the web. Since this bias cannot be completely avoided, we argue that our estimate is a lower bound on the number of privacy policies on the web.

Based on the PrivaSeer Corpus, we found that $n_1 = 1,005,380$. Based on the CoLIPPs Corpus, we found that $n_2 = 612,046$ and $m = 206,417$. Using equation (4.1), we estimated that the lower bound on the total number of privacy policies on the web is 2,981,047. Using equation (4.2), we estimated that the 95% confidence interval on the estimate is 9,333.

4.5.1 Top Level Domain Analysis

In addition to estimating the lower bound of the number of privacy policies on the web, we estimated the number of privacy policies in each top level domain (TLD). Similar to our earlier estimate, we counted the number of common privacy policies between the PrivaSeer Corpus and the CoLIPPs Corpus based on the domain name and separated them by TLD. We then applied the Lincoln-Petersen technique to estimate the number of privacy policies in each TLD. Table 4.2 shows the top 10 TLDs with the largest number of estimated privacy policy documents. In this table, *Overlaps* and *Estimate* refer to the number of overlaps between the PrivaSeer Corpus and the CoLIPPs Corpus and the total estimated count of URLs in each TLD respectively. From this table, we can see that .com domains have the overwhelming majority of privacy policies, followed by .uk and .org domains. We also note that a number of country level TLDs, such as .uk, .au, .ca, .ie, .de, and .in contain a large number of privacy policies. A number of TLDs that appear in the top 10 registered TLDs on the internet (Verisign, n.d.)

such as .tk, .cn, .icu, .nl and .ru do not appear in the top ten TLD with the greatest number of privacy policies. This could be due to the fact that we only capture English language privacy policies while a large percentage of websites in .cn, .nl and .ru TLDs are likely in Chinese, Dutch and Russian respectively. The differences in privacy policy frequencies in various country level TLDs could also be due to the privacy laws in the respective countries. Additionally, about 37% of the total number of registered domains on the internet have a .com TLD, however, 63% of the English language privacy policies are contained within .com TLD. This phenomenon might be explained by the fact that commercial establishments might prefer to host their websites on the .com TLD.

Figure 4.2 shows the TLDs with the highest number of English language policies. Each bar in the figure depicts the percentage of domains with and without a privacy policy for that TLD. Here we can see that country level TLDs with predominantly English speaking population have the highest ratio of domains with policies to domains without policies, namely .uk (United Kingdom) and .au (Australia) and .ie (Ireland). We can see that .de, the German TLD contains a very low ratio of domains with policies to domains without policies, providing further evidence to our hypothesis. We see the .in (India) TLD in spite of having a large population of English speakers and websites in English contain a much smaller ratio of domains with policies to without policies. This is likely due to the fact that unlike USA and the member states of the EU, India does not have a well defined law requiring privacy policies on websites.

Table 4.2 can also be used to identify bias while estimating population sizes. By breaking up the estimates into component categories (Pollock et al., 1990) and analysing TLD estimates separately, we can identify instances of bias in the data. For example, we see TLD .info has a large difference between the number of overlaps and estimates. This might be due to the fact that one of the corpora disproportionately captured instances that TLD, thereby causing a positive estimation bias, i.e., overestimation of the number of privacy policies. The CoLIPPs Corpus was created from URLs in LinkedIn. It is reasonable to assume that not many companies would host their website on a .info domain. Thus, .info being underrepresented in the CoLIPPs Corpus might lead to an artificially low number of overlapping URLs between the corpora, causing an overestimation of the number of privacy policies for the .info TLD. We can also see that TLDs which make up a large percentage of the total estimate such as .com, .uk and .org do not show any obvious signs of bias.

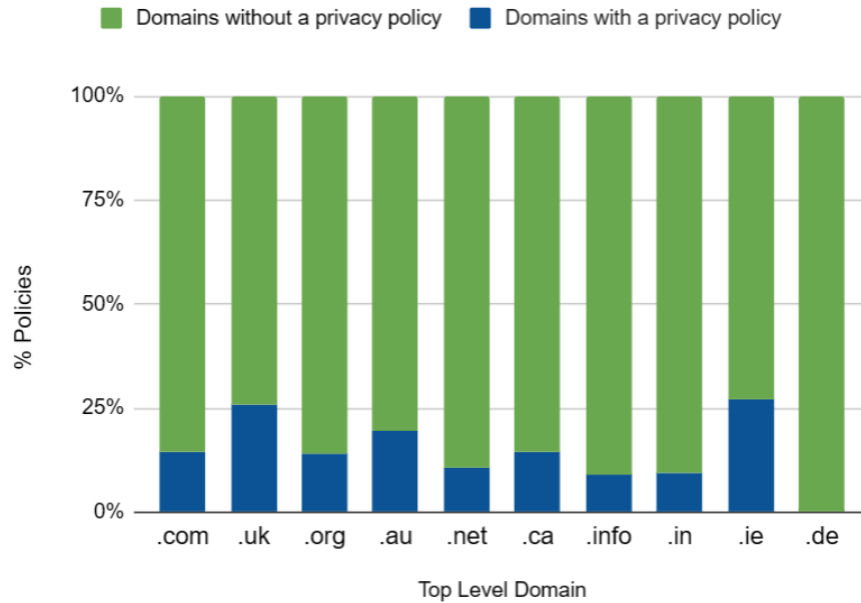


Figure 4.2: Comparison between the %domains containing a privacy policy and %domains not containing a privacy policies for the top ten TLDs with the most English language policies

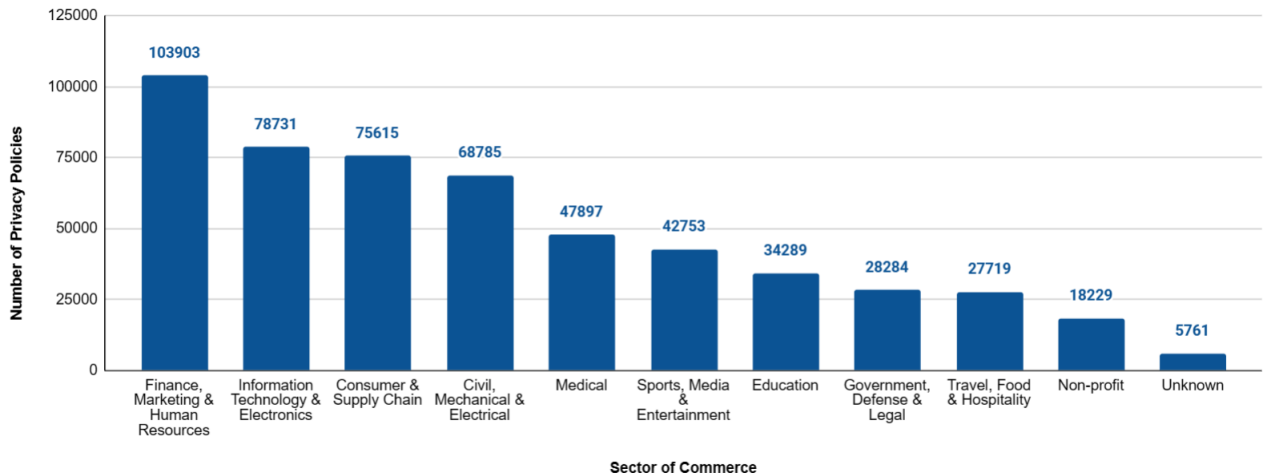


Figure 4.3: Distribution of the number of privacy policies in each sector of commerce in the CoLIPPs Corpus

4.5.2 Sector of Commerce Analysis

The sector of commerce is an important factor to consider in the context of privacy policies. Not only do different sectors of commerce differ in trends in privacy practices and policy availability, they also collect different categories of personal information types from the user. For example, financial institutions might typically require a user's social

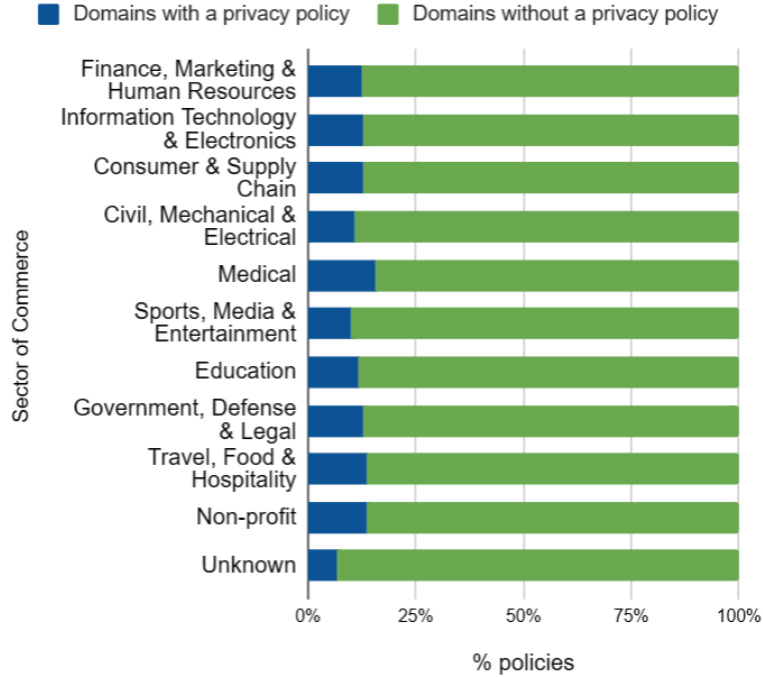


Figure 4.4: Comparison between the %domains containing a privacy policy and %domains not containing a privacy policies for all sectors of commerce

security number, but this is not the case for e-commerce companies. The Free Company Dataset, from which we built the CoLIPPs Corpus, maps company websites to a set of 148 unique industries. The company categorization scheme employed in the Free Company Dataset comes from the categorization scheme followed by LinkedIn (companies self-report the industry to which they belong). We consolidated the 148 industries categories into 11 sectors of commerce. We make both the sector of commerce labels and industry labels available as part of the CoLIPPs corpus metadata.

Figure 4.3 shows the count privacy policies in each sectors of commerce in the CoLIPPs Corpus. We can see that the sectors *finance, marketing and human resources* and *information technology and electronics* having the greatest segments with 17% and 14% of the privacy policies respectively. Figure 4.4 shows the percentage of domains containing a privacy policies and the percentage of domains not containing a privacy policy for all the sectors of commerce. From the figure we can see that the medical sector has the highest ratio of domains containing policies to those that do not. On the other hand, the sports media and entertainment sector has the lowest. We also see that there is a general trend that consumer facing sectors that could potentially serve as a rich source of user data such as IT, marketing, hospitality, and consumer sectors have a higher percentage of domains that have policies. This is true for sectors that could

potentially handle sensitive data as well such as medicine, finance, human resources, and non-profit sectors. On the other, non-consumer facing sectors such as civil, mechanical and electrical as well as government and defence have the lowest percentage of websites with policies. However the *sports, media and entertainment* sector which potentially could serve as a rich source of user data has one of the lowest percentage of domains with policies, potentially indicating an issue with policy availability.

4.6 Discussion

A user's journey to find a typical website's privacy policy would generally involve clicking on the privacy policy hyperlink on the website's landing page. However, based on our estimate, the chance that the user would find a privacy policy URL on the website is only about 34%. If the user does find the URL, we estimate that there is a 1.37% chance that the link leads to an error page. If the user is successfully directed to the privacy policy page by clicking on the link, we estimate that there is 1.23% chance that the privacy policy has no content (on an English language website). Further, if the user is on a website that is offered in multiple languages, we estimate that there is a 41.9% chance that there is a mismatch between the language in which the privacy policy is offered and the user's choice. If the user finally accesses the privacy policy without encountering any of the above failure modes, they still must spend a considerable amount of time trying to understand the privacy policy.

The General Data Protection Regulation (GDPR) is a regulation in European Union law on data protection and privacy. The GDPR has been hailed as a landmark in privacy protection law and has inspired a number of other user privacy laws around the world. According to the GDPR, entities that collect personal information from its users need to provide notice regarding what information they are collecting and why they are collecting it. *Notice and Choice* is one of primary rationales behind the GDPR and thus works only when privacy policies are readily available and easily readable. However, the lack of widespread availability of privacy policy documents online provides a basic obstacle to notice and choice. It is therefore concerning that only about 34% of websites provide a privacy policy. We acknowledge the fact that some websites that don't post a privacy policy might indeed not collect any personal information from its users. However, there is no way for the user to know whether a website collects any information or whether they are simply unable to find the policy if the company has not posted a policy on its website. Additionally, most web servers automatically log server traffic data that

includes user IP address, pages visited and time of visit. Websites might also have integration with third party services that might in-turn collect user information. It is therefore injudicious to assume that companies that have not made privacy policies available on their websites simply do not include any third-party integration and have disabled automatic user data collection features. To further complicate the problem, the nature of regulation enforcement can also pose a serious challenge since some jurisdictions (such as the European Union) rely on user complaints to investigate privacy practices (Supervisor, 2018) while others (such as the United States) only investigate when a privacy policy is at odds with real world privacy practices. We acknowledge that some websites that don't post a privacy policy might be present in jurisdictions that do not require one. However, users from parts of the world with different privacy expectations might be able to access websites that are not legally obligated to post a privacy policy. We therefore argue that posting a privacy policy should be a standard practice for any entity or organisation.

The availability of privacy policies and the ability of users to understand them are fundamental to ensuring that individuals can make informed decisions about their personal information. However, the lack of standardization and the sheer volume of privacy policies on the web can create significant barriers to access and understanding. This work contributes to ongoing efforts to promote transparency and accountability in on-line data privacy practices, which are critical to the continued growth and development of the digital economy. In particular, this research provides important insights into the current state of privacy policy practices on the web, including an estimate of the total number of privacy policies, and the types of organizations that are most likely to have privacy policies in place. These findings can inform efforts to develop more effective privacy policy standards and best practices, as well as to improve the accessibility and comprehensibility of existing policies for users. Additionally, this work can help policymakers and industry stakeholders better understand the challenges and opportunities associated with online data privacy practices. For example, in our study, we found that while websites that could potentially serve as a rich source of user data are likely to contain a privacy policy, websites in the *sports, media and entertainment* sector contain a unusually low percentage of websites with policies. As data privacy regulations continue to evolve and expand globally, it is essential that researchers and practitioners in the community stay up-to-date with the latest trends and best practices in this area. By contributing to this body of knowledge, this research can help support the development of more effective policies and practices that protect individual privacy rights while

also facilitating innovation and growth in the digital economy. The issues regarding the availability of privacy policies is a multi-faceted problem. Thus, combined effort from regulators and researchers in multiple disciplines is required to improve trends in user data protection online.

4.7 Conclusion

Online privacy policies are crucial in safeguarding users’ personal data and privacy rights. However, obtaining these policies can be a challenging task. In this study, we analyzed the online privacy policy landscape and studied the unavailability of privacy policies derived from a large dataset of company domains. At various stages of document collection and classification, we encountered a number of failures in obtaining policies and estimated frequencies of unavailability. The different failures include broken links, language inconsistencies between the document and its landing page, placeholder texts, and empty content on the privacy policy document page. We publish this novel corpus, containing 600k privacy policy corpus (with rich metadata) to the research community for further analysis.

Using a *capture-recapture technique*, we established a lower bound estimate of 2,981,047 for the total number of English language privacy policies available on the web. This result also highlights the effectiveness of multidisciplinary approaches in the field of privacy. By analyzing the distribution of privacy policies across various top-level domains (TLDs), we found that the majority of English privacy policies on the web are hosted on domains with .com, .uk, and .org TLDs. Furthermore, we demonstrated that the analysis of the distribution of privacy policies in TLDs can be leveraged to make predictions about the efficacy of privacy regulations in specific jurisdictions. Notably, our analysis revealed an unusually low number of .in domains with privacy policies in comparison to other English-speaking country-level TLDs such as .uk, .ie, .au, and .ca. We attribute this finding to the absence of national privacy regulations requiring the use of privacy policies in India. Additionally, our examination of the distribution of policies across different sectors of commerce suggests that domains within the education, sports, media, and entertainment sectors are more likely to not contain privacy policies than domains in other sectors.

Future work can explore differences and similarities of privacy policies across domains and industries while comparing instances of their failure modes. Studying failure modes based on industry regulations as well as jurisdictional regulations can further illuminate

the online privacy policy landscape and therefore enable researchers and regulators to draw meaningful insight regarding privacy policies on the web.

Chapter 5 —

Privacy policy search engine

5.1 Background

Without automated tools, regulators in some jurisdictions (such as the European Union) rely on user complaints to investigate privacy practices (Supervisor, 2018) while others (such as the United States) rely on organisations to self-certify their compliance¹ and only investigate when a privacy policy is at odds with real world privacy practices.

To remedy the lack of a publicly accessible large-scale privacy policy resource, we present PrivaSeer², a privacy policy search engine that currently indexes over 4 million privacy policies collected from the web as described in Chapter 3. However, the numbers here indicate indexing only from the first two crawls which encompass roughly 1.4 million privacy policy documents. PrivaSeer can be used to find policies based on policy text using facets such as sector of commerce, policy vagueness, policy readability, tracking technology mentioned, regulatory bodies mentioned and regulations or cross-border agreements mentioned in the policy text. Search results can be ranked by popularity of the website of the policy, relevance based on the query and the probability that a document is a privacy policy. To the best of our knowledge, PrivaSeer is the first search engine specifically designed to support privacy research.

PrivaSeer gives researchers the ability to quantify and examine sets of policies based on key features, enabling them to discover trends in privacy practices online. Similarly, policy regulators and legal experts can use PrivaSeer to find anomalies in policies, thereby empowering them to protect users from privacy-eroding practices. The simple and intuitive search interface allows privacy concerned web users to find features of particular privacy policies and search for privacy-friendly alternatives for everyday

¹<https://www.privacyshield.gov/Program-Overview>

²<https://privaseer.ist.psu.edu/>

services.

5.2 Related Work

The related work can be categorised into two areas: collections of privacy policies and methods to simplify privacy policy documents.

To the best of our knowledge, PrivaSeer is the first privacy policy search engine, but a few prior attempts have focused on making privacy policies more accessible to the public. The Usable Privacy Policy Project made available a collection of 115 privacy policies with fine grained human annotation of privacy practices in policies (Wilson et al., 2016). These policies can be accessed through the website and filtered based on the URL³. They display the sector of activity, readability and popularity of the website from which the policy was originally obtained. In addition, they also created a collection of seven thousand privacy policies which contain machine annotated privacy practices which can be filtered by URL. Similarly, Polisis⁴ is a collection of about 31,000 privacy policies which generates automatic summaries of privacy policies based on various data practices. While the privacy policies can be filtered based on their URL, they cannot be searched based on user queries (Harkous et al., 2018). More recently, Amos et al. (2020) released a longitudinal corpus of privacy policies collected from around 130,000 websites.

5.3 Search Interface

The user interface of PrivaSeer resembles a standard search engine, in order to keep the system familiar and easy to use. A user enters a query in the search text box on the landing page and can opt to search either the privacy policy URLs or the policy text by selecting a radio button. Figure 5.1 shows a screenshot of the landing page. Clicking *Search* takes the user to the results page.

By default, the privacy policies in the results page are ordered based on a custom ranking function discussed in Section 6. The result page displays the top ten results with options to go to the next page. Each result has the title of the webpage, the URL, the date it was crawled and snippets of text in the document with words matching the query words highlighted. The user can re-filter the results based on search facets on either side of the page. Figure 5.2 shows a screenshot of the results page for the query ‘address’.

³<https://explore.usableprivacy.org/>

⁴<https://pribot.org/polisis>

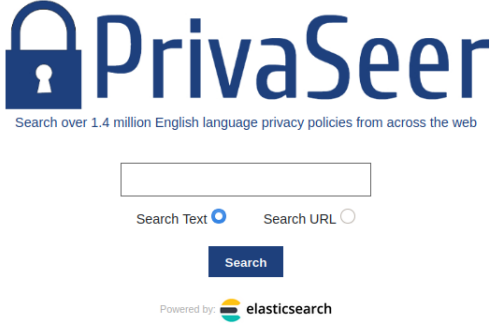


Figure 5.1: Snapshot of landing page

a common personal information type mentioned in privacy policies. In the left hand column, the facets *sector of commerce*, *tracking technology*, *self-regulatory bodies* and *regulations and agreements* are present. Users can use the check-boxes to select one or more facets in each category. In the right hand column, users have the option to change the ranking technique to rank by either relevance based on the query, PageRank or the default custom ranking by using the *sort by* drop-down menu. They can also filter the search results based on Flesch Kincaid reading level (Kincaid et al., 1975) by entering readability levels between 0 and 18. Finally, they can filter the results by the percentage of vague sentences in the policy by entering a decimal between 0 and 1. Section 7 further describes each facet category.

5.4 Indexing

We created two separate indexes: one for privacy policy web pages with non-policy content included and one without. Non-policy content refers to content in a privacy policy web page such as header, footer and navigation menus which are irrelevant to the privacy policy as a legal text. We used Elasticsearch (Gormley & Tong, 2015) to create an inverted index and divided the documents into the title, URL, and body for indexing. We tokenized the body and title of the privacy policy using grammar based tokenization that works based on the Unicode text segmentation algorithm (Davis & Iancu, 2012) and tokenized the URL based on a regex tokenizer. PageRank, length, readability and vagueness were indexed as numeric types while industry, tracking technology, regulations/agreements and self-regulatory bodies were indexed as text fields, either singly or as an array in cases where a privacy policy could contain multiple types of a single facet.

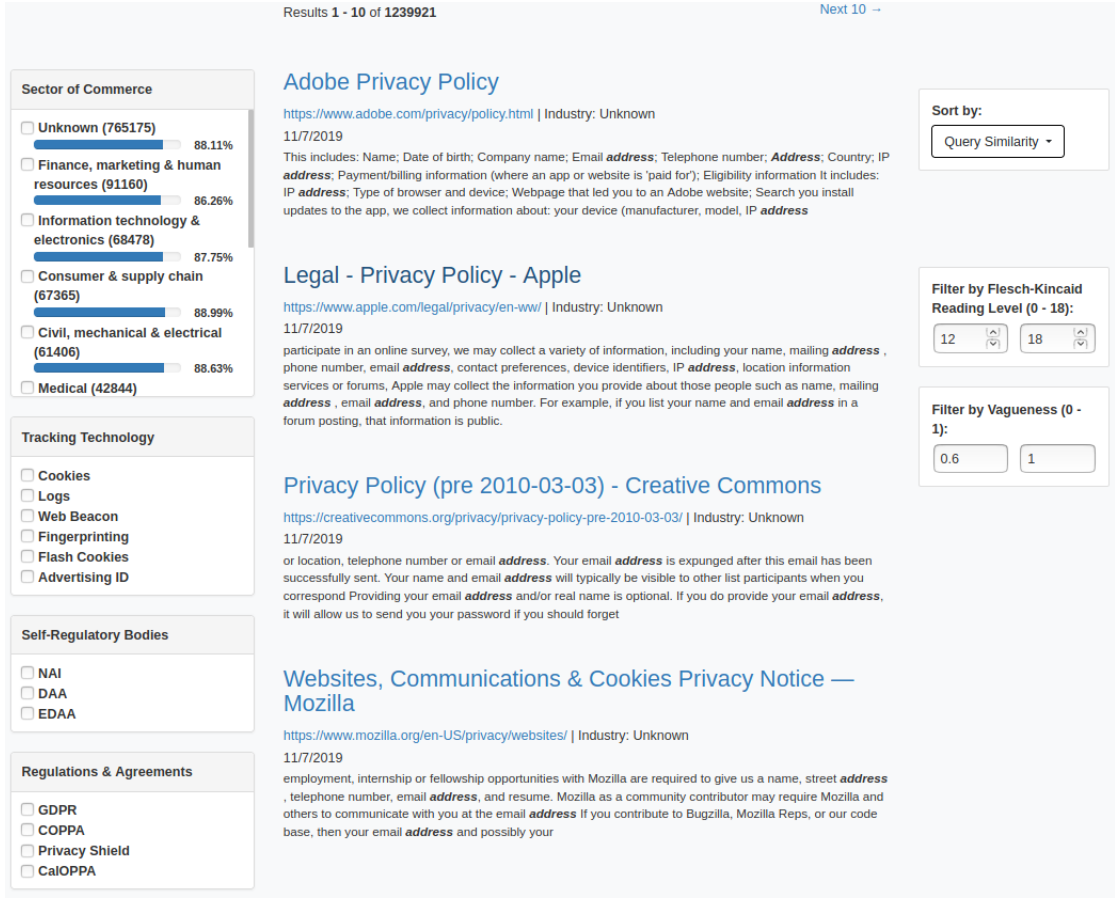


Figure 5.2: Snapshot of results page

5.5 Ranking

The results in PrivaSeer are ranked based on PageRank, query based document relevance and the probability of the document being a privacy policy.

PrivaSeer uses the bag-of-words based Okapi BM25 (Robertson et al., 1999) ranking function to estimate the relevance of a document given a search query. Given a search query Q with terms q_i where $i = 1 \dots n$, the score of a document D is given by the following function.

$$\sum_1^n idf(q_i) \times \frac{(k_1 + 1).tf(q_i, D)}{tf(q_i, D) + k_1(1 - b + b \cdot |D|/dl_{avg})} \quad (5.1)$$

Where, $idf(q_i)$ is given by the following equation.

$$idf(q_i) = \log \frac{N - n(q_i) + 0.5}{n(q_i) + 0.5} \quad (5.2)$$

In the equations, N is the total number of documents in the collection, $n(q_i)$ is the number of documents containing q_i , $tf(q_i, D)$ is the term frequency of q_i in document D , k_1 and b are tuned constants, $|D|$ is the number of words in document D and dl_{avg} is the average document length in the collection.

We extracted the PageRanks of the domains in the corpus Common Crawl’s web graph. Common Crawl use the Gauss-Seidel algorithm (Arasu et al., 2002) to calculate the PageRanks in the web graph. Only a few domains have a high PageRank suggesting that ranking based only on PageRank might limit the discovery of privacy policies from the domains that are not very popular. The custom ranking function combines the scores derived from query based document relevance with PageRank and the probability of the document being a privacy policy. The final score of a document D given query Q is given by the following function.

$$P(D) \times Relevance(D, Q) \times \log_{10}(D_{pr}) \quad (5.3)$$

In the equation, $P(D)$ is the probability that the document is a privacy policy, $Relevance(D, Q)$ is defined in equation 1, and D_{pr} is the PageRank of the website from which the document was crawled.

The PrivaSeer Corpus was created by training a random forest model to classify whether a document is a privacy policy. Srinath et al. (2020) labeled 1000 crawled documents as either a privacy policy or not. We used the labeled data to train a machine learning model and obtained the probability of a document being a privacy policy. We used 100 documents as the validation set to tune hyperparameters and divided the rest of the documents into train and test sets in the ratio 4:1. We then tokenized and removed stop words before using term-frequency inverse-document-frequency features extracted from the URL and document. The average precision and recall score after 5-fold cross validation were 0.96 and 0.97 respectively.

Figure 5.3 shows the distribution of the probabilities of documents classified as a privacy policy for all the documents in the PrivaSeer Corpus. The horizontal axis begins at 0.5 since the binary classification cutoff probability was 0.5, with 1 being the label for a privacy policy and 0 being the label for its negation. The figure shows that most of the documents are classified with high confidence with only a few documents having a probability less than 0.7.

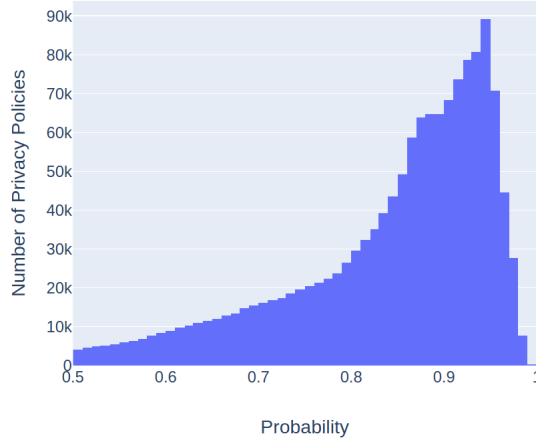


Figure 5.3: Distribution of probabilities of documents (being a privacy policy) in the PrivaSeer Corpus

5.6 Filtering and Observations on the Document Set

5.6.1 Sectors of Commerce

The Free Company Dataset, which was used to obtain seed URLs for the PrivaSeer Corpus, maps website URLs to a set of 148 unique industries. Two researchers worked independently and arrived at a consensus to consolidate the industries into 11 sectors of commerce. Table 5.1 shows the distribution of privacy policies across different sectors of commerce in the PrivaSeer Corpus. *Unknown* consists of extracted privacy policies whose sector of commerce information could not be found on the Free Company Dataset. Expected norms for privacy practices differ based upon sectors of commerce. For example, privacy policies in the medical sector are more likely to address users’ health information, which has its own privacy laws (i.e., HIPAA in the US). Thus, we provide sector of commerce as a filter facet to enable policy regulators to compare and find anomalies between privacy policies of the same sector.

Figure 5.2 shows the screenshot of the results page with sector of commerce as a filter facet. For a given query, the number of privacy policies in each sector is specified next to sector name. The progress bar and percentage value for each sector indicate the number of privacy policies retrieved for the query out of the total number of privacy policies for that sector in PrivaSeer Corpus.

Sector of Commerce	Number
Unknown	858,395
Finance, Marketing and Human Resources	106,732
Information Technology and Electronics	82,192
Consumer and Supply Chain	77,477
Civil, Mechanical and Electrical	70,209
Medical	49,918
Sports, Media and Entertainment	43,912
Education	35,468
Government, Defense and Legal	29,037
Travel, Food and Hospitality	28,290
Non-Profit	18,688

Table 5.1: Distribution of privacy policies across different sectors of commerce

5.6.2 Readability

Readability of a text can be defined as *the ease of understanding or comprehension due to the style of writing* (Klare et al., 1963). One of the main critiques of privacy policies is that they are too complicated to read and understand. Studies have found that privacy policies are difficult to read and require a college-level reading ability (Fabian et al., 2017) (Ermakova et al., 2015). The online privacy paradigm follows the *Notice and Choice* framework. *Notice* is a presentation of terms by an organisation, usually in the form of a privacy policy and *choice* is an action by a user signifying the acceptance of terms (Sloan & Warner, 2014). When privacy policies are difficult to understand, the notice and choice framework breaks down. To assess readability, we calculate the Flesch-Kincaid Grade Level (FKG) (Kincaid et al., 1975) for all the policies in the corpus and include it as a facet to filter privacy policies. FKG gives the United States school grade level an average user would need to be in order to understand the text. Srinath et al. show the distribution of readability scores in the PrivaSeer Corpus based on a number of readability techniques (Srinath et al., 2020).

5.6.3 Tracking Technology

Tracking technologies are used by organisations to keep track of web users’ browsing habits. We selected six different types of tracking technologies and extracted their mentions in all the policies in the PrivaSeer Corpus. Table 5.2 shows the distribution of these mentions, extracted using regex queries in an approach similar to Amos et al. (2020). To study the effectiveness of the regex technique, we manually sampled 10

privacy policies from each category of the facet and found no false positives for any category. Intuitively, privacy policies rarely mention tracking technologies which they do not use. While there is a possibility that mentions of some tracking technologies were not captured by the regex technique, thereby leading to false negatives, we believe that the regex expressions captured the common terms for all tracking technology thereby minimizing false negatives.

Studies have found a misalignment between the use and mentions of various tracking technology in privacy policies (Amos et al., 2020). While tracking technologies are common in practice, they may not always be mentioned in the privacy policy. Thus, to enable further investigation of discrepancies and trends in the use of tracking technology we include tracking technologies as a facet.

Tracking Technology	Number of Policies	% of Total
Cookies	1,179,351	84.2%
Logs	249,901	17.8%
Web Beacon	236,099	16.9%
Fingerprinting	73,969	5.3%
Flash Cookies	39,199	2.8%
Advertising ID	15,366	1.1%

Table 5.2: Distribution of Tracking Technology

5.6.4 Self-Regulatory Bodies

Some jurisdictions rely on organisations to self-certify their privacy regulation compliance. Organisations therefore work with self-regulatory bodies to provide them with privacy seals and certificates verifying the organization’s adherence to certain specified privacy standards (Rodrigues et al., 2013). Amos et al. presented a longitudinal analysis of self-regulatory compliance by examining the mentions of self-regulatory bodies in privacy policies (Amos et al., 2020). We applied the same method of using regex queries to extract mentions of nine self-regulatory bodies in privacy policies of the PrivaSeer Corpus. Similar to the regex technique applied to extracting tracking technologies, we sampled ten random privacy policies for each item in the facet and found no false positives.

Table 5.3 shows the percentage of privacy policies mentioning each of the self-regulatory organizations in the PrivaSeer Corpus. Only initiatives such as Network Advertising Initiative (NAI), Digital Advertising Alliance (DAA) and European Inter-

Table 5.3: Distribution of Self-Regulatory Bodies

Self-Regulatory Bodies	Number	% of Total
NAI	88,964	6.35%
DAA	72,754	5.19%
EDAA	22,874	1.63%
BBBOnLine	3,190	< 1%
TrustArc	2,300	
CNIL	1,767	
ePrivacy	899	
VeraSafe	180	
Evidon	109	

active Digital Advertising Alliance (EDAA) that develop self-regulatory standards for online or digital advertising, have significant number of mentions (present in over 1% of privacy policies in the corpus). Therefore, we only provide these as filterable items in PrivaSeer.

5.6.5 Regulations and Agreements

While some jurisdiction rely on organisations to self-certify their privacy compliance, others rely on concrete regulations and cross-border agreements. Similar to self-regulatory bodies, we extracted mentions of eight regulations and cross-border agreements in privacy policies of the PrivaSeer Corpus. We also included a sector-specific government regulation, the Health Insurance Portability and Accountability Act (HIPAA). Table 5.4 shows the percentage of privacy policies mentioning different regulations and agreements in the PrivaSeer Corpus. GDPR and Privacy Shield have the highest number of mentions among regulations and cross-border agreements respectively. We include regulations and agreements as a filter facet in PrivaSeer to enable users to identify privacy policies that refer to these regulations.

5.6.6 Vague Language

A term is regarded as *vague* if it admits borderline cases, where speakers are reluctant to say either the term definitely applies or does not apply (Bhatia et al., 2016). Vagueness in privacy policies is a pervasive problem (Lebanoff & Liu, 2018), limiting the ability of readers to precisely interpret their contents. Uncertainty in future needs prompts organisations to resort to using vague language to describe their privacy practices (Bhatia

Table 5.4: Distribution of Regulations & Agreements

Regulations & Agreements	Number	% of Total
GDPR	228,726	16.33%
COPPA	73,745	5.27%
Privacy Shield	62,778	4.48%
CalOPPA	57,819	4.13%
CCPA	13,215	< 1%
SCC	6,834	
HIPAA	3,713	
BCR	2,105	

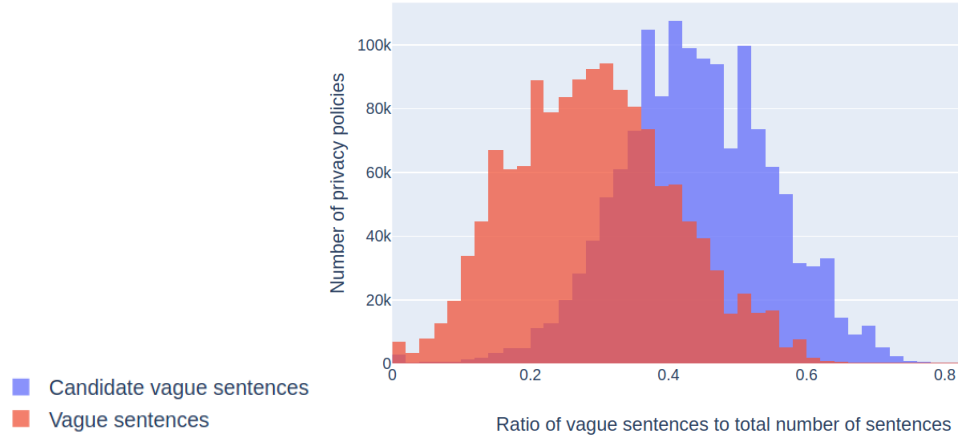


Figure 5.4: Distribution of vague sentences

et al., 2016). This diminishes the effectiveness of policies making them unclear to users, thereby reducing trust and causing potential user privacy issues. Thus, we calculate the vagueness scores of all policies in the PrivaSeer Corpus and make them available as a search facet, for regulators and researchers to study at scale.

We use the corpus on vagueness in privacy policies made available by Lebanoff and Liu (Lebanoff & Liu, 2018) to calculate the vagueness of policies in PrivaSeer. To create their corpus, Lebanoff and Liu first extracted sentences from 100 privacy policies that contain 40 cue words for vagueness (Bhatia et al., 2016). Each sentence was considered separately and the vague words/phrases in it were identified and annotated. Since the sentence context was not considered, co-referential words were annotated as vague. For example, in the sentence *You can find out more about this on the anonymous edits page*, the word ‘this’ was annotated as vague. Since our aim was to find the vagueness of the privacy policy as a whole, we ignored annotations on words that were only annotated as vague due to co-reference issues. We ignored the annotations on the following words

when they were annotated as vague: *it, this they, them, that, these, here, there, you, us, we, and following*. We also ignored annotations on the following phrases when they were annotated as vague: *personal information, personally identifiable information, and third party (parties)* as they might have been defined in the privacy policy prior to usage.

We treated the problem as a token classification problem where each word in a sentence would be predicted as either vague or not. We fine-tuned a pretrained transformer based model, namely Roberta (Y. Liu et al., 2019), using the Roberta token classification head from Huggingface (Wolf et al., 2020). We divided the corpus into train, development, and test sets in the ratio 3:1:1. We used the development set for hyper-parameter tuning. Table 5.5 shows the results for vague word prediction. While Lebanoff and Liu achieve better precision and recall scores, the corpus that we report on is a modified version due to the removed co-references.

For predicting vagueness of all the policies in the corpus, we extracted sentences from the corpus that had any one of the 40 cue words (Bhatia et al., 2016) similar to Lebanoff and Liu (Lebanoff & Liu, 2018). We call these *candidate vague sentences*. Following the candidate sentence extraction, if any word in a sentence was found to be vague by the Roberta model, we considered the sentence to be vague. We then normalised the number of vague sentences with the total number of sentences in the privacy policy to obtain a vagueness score for each policy.

Table 5.5: Vague Word Prediction Results

Model	Precision	Recall	F1
Lebanoff and Liu	68.4	53.8	60.08
Roberta	65.1	52.6	58.3

The distribution of candidates and vague sentences is shown in Figure 5.4. From the figure we can see that on average around 50% of the sentences in a privacy policy are candidates for vague sentences, while about 30% are actually vague. There appears to be a long tail with some privacy policies having almost no vague sentences and some with almost all vague sentences. Manual evaluation of the policies in tail shows that most of them are very short with at most three or four sentences. Figure 5.2 shows the filter facet for vagueness based on a measure of the ratio of vague sentences to the total number of sentences in the policy. Users can filter results by entering a range between 0 and 1 to select the proportion of vague sentences they would like to see in policies.

5.7 Ranking Evaluation and Discussion

We perform an exploratory evaluation, since no prior work exists on evaluating a privacy policy search engine. Precision at k or $P@k$ measures the number of relevant results among the top k returned results. We report precision at 10 and precision at 5 scores for two indexes of privacy policies as discussed in Section 5, one with the context provided by non-policy content and one without.

Prior research identified ten categories of privacy practices that lawyers expect privacy policies to contain (Wilson et al., 2016). To evaluate PrivaSeer, we created three themes for queries based on the ten categories in prior work. These themes comprise of *personal information type (PI)*, *security information (S)*, and *privacy practice type (PP)*. The queries were designed and evaluated so that even if a returned result for a query was a privacy policy with the query words, it was deemed irrelevant if it did not fall in the expected category. For example, The query ‘health information’ is from the category *personal information type*. If a returned result for the query was a privacy policy from a hospital which did not mention how users’ health information would be collected or managed, then the query was deemed irrelevant.

Table 5.6: Queries and their categories

Category	Queries
Personal Information Type	payment information, health information, social security number, phone number, photos, private messages, microphone
Security	firewall, encryption, SSL, data breach, deletion
Privacy Practice Type	opt-out, retention period, change notification, do not track, European audience

The categories of queries and each query that was used for evaluation are shown in Table 5.6. Table 5.7 shows the comparison of $P@5$ and $P@10$ results between three ranking schemes over the different type of query categories and indexes. The results suggest that the custom ranking technique works best followed by PageRank and finally the simple query-document matching.

We tested a version of the custom ranking technique without document probability scores and found that the results were slightly better than either PageRank or query based document relevance individually. Although this technique was able to leverage PageRank and query based document relevance scores together, we found that it per-

Table 5.7: PrivaSeer evaluation results

	Non-policy content Excluded						Non-policy content Included					
	PI		S		PP		PI		S		PP	
	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10
Relevance	0.54	0.48	0.28	0.38	0.66	0.63	0.37	0.31	0.48	0.44	0.6	0.62
PageRank	0.6	0.56	0.6	0.62	0.48	0.5	0.51	0.41	0.56	0.6	0.72	0.7
Custom	0.88	0.9	0.92	0.92	1	0.9	0.83	0.76	0.76	0.78	0.9	0.9

formed poorly in cases where false positive privacy policies came from domains with a high PageRank. It was only able to perform well when both PageRank and query based document relevance scores presented reasonable results on their own. The use of document probabilities significantly improved ranking performance. The document probability scores suppress documents with a high PageRank or high query based document relevance scores but which might not be a privacy policy in reality.

Performance of all the techniques deteriorated on the index with non-policy content, across all the categories. This suggests that content in the header, footer or navigation menu do not provide much context while ranking queries related to privacy practices. It is likely that non-policy content would improve ranking in cases where users would like to filter results based on a specific industry. While the ‘sector of commerce’ facet allows users to filter results based on coarse grained industry categories, queries which include industry specific words on an index with non-policy content might serve as a stronger filter.

The custom ranking technique outperforms the PageRank and query based document relevance techniques and also has a higher variability in the returned results when compared to the PageRank technique. The PageRank technique usually returns the same set of documents for most queries. We hypothesize that this behaviour is because most popular websites have a comprehensive coverage of privacy practices.

5.8 Conclusion

We present PrivaSeer, the first privacy policy search engine. PrivaSeer is a necessary tool that is the first of its kind and is helpful to several distinct groups with goals in furthering user privacy. Documents can be ranked by query based document relevance scores, PageRank values, and document probabilities. They also can be filtered based on sector of commerce, policy vagueness, policy readability, tracking technology mentioned,

regulatory bodies mentioned, and regulations/cross-border agreements mentioned in the policy text.

On average about 30% of the sentences in a privacy policy were found to have at least one vague word in them. This suggests that vagueness in privacy policy documents is a pervasive problem. We used regex text matching to extract details about tracking technology, regulatory bodies, and regulations/cross-border agreements and found non instances of false positives. We believe this is because privacy policies only record elements of privacy that they use/comply while rarely mentioning other elements/alternatives that exist.

An exploratory evaluation of PrivaSeer based on PageRank, query based relevance, and our custom ranking technique found that the custom ranking technique outperformed the others in all categories. We found that our custom technique had higher variability in returned results and was able to overcome limitations caused by the presence of false positive privacy policies in the results. Future work could concentrate on adding a temporal component to the collection of privacy policies and explore alternative ranking methods.

Chapter 6 —

Approaches to Simplify Privacy Policies

6.1 Background

Do internet users care about their online privacy? While studies have shown that users care about their privacy online (Spiekermann et al., 2001), they are also willing to give away their personal information (Barnes, 2006). Studies have found that this is often because, *privacy policies*, which are intended to inform users about what happens with their data online, are notoriously difficult to understand and time-consuming to read (Obar & Oeldorf-Hirsch, 2018). An average user needs to spend ~ 200 hours to read all the policies that they come across each year (McDonald & Cranor, 2008). Moreover, privacy policies are written at a college reading level (Ermakova et al., 2015), use complicated legal jargon and are hard to comprehend (Meiselwitz, 2013).

Some studies have attempted to ease the design of privacy policies (Schaub et al., 2015) to make them more comprehensible to users. Kelley et al. (2010) found that presenting privacy information visually in the form of a *privacy nutrition label* significantly improved the ability of users to find and understand privacy information. While companies such as Apple and Google have incorporated a *privacy label* as a way to systematically present users with concise summaries of an app’s data practices, they often fail to answer important privacy questions (Zhang & Sadeh, 2023). Moreover, most organizations do not provide access to easy-to-read labels. Given that there are several issues related to conducting privacy policy research (Mhaidli et al., 2023), scaling the creation of such labels using NLP approaches is a non-trivial task.

We hence propose to automatically present users information regarding their online privacy. To this end, the undertake two tasks to automatically extract information from privacy policies.

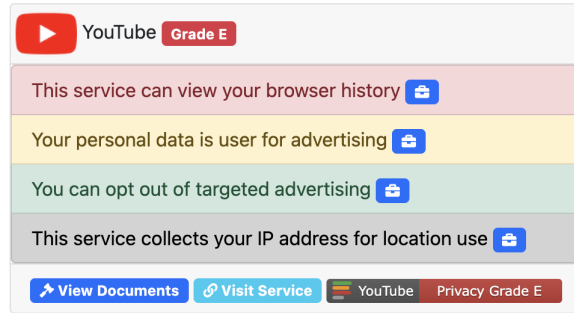


Figure 6.1: Snapshot of annotated ToS;DR *cases* and assigned privacy letter grade for YouTube

1. Automatic extraction of Data Practices from Privacy Policies
2. Automatic Extraction of Dates from Privacy Policies

6.2 Automatic extraction of Data Practices

6.2.1 Methodology

6.2.1.1 Dataset

Terms of Service; Didn't read (ToS;DR) aims to make understanding privacy policies easier by crowdsourcing annotations on them (Roy et al., 2012). On the platform, internet users sign up to annotate policies by matching a policy excerpt to a pre-defined data practice (called a “case” on the platform). Once an annotator makes an excerpt-case match, a ToS;DR moderator either accepts the annotation or rejects it with comments. When a threshold percent of moderator approved *case*-excerpt matches are annotated for a policy, it is given a letter grade.

ToS;DR comprises 130 privacy policy-related *cases*. A case on ToS;DR contains an average of 83 policy excerpts (min: 2; max: 83), each from a different service. Each excerpt has an average length of 32.26 words/excerpt. Some *cases* describe similar but contrasting data practices i.e. different approaches to handling the same data type. We deemed these as **contrasting cases**. Following is an example of two contrasting cases:

‘Your personal data is not sold’

‘Your personal data is sold unless you opt out’

Two authors manually evaluated the cases in ToS;DR and grouped *contrasting cases* into 24 clusters (min: 2; max: 6 excerpts/cluster). We note that *contrasting cases* might often be a negation of one another, or might differ by describing a special circumstance of a general case, such as,

‘*You can delete your content from this service*’

‘*You cannot delete your content from this service*’

‘*You cannot delete your content, but it makes sense for this service*’

The remaining 67 *cases* had no contrasting data practices. We deemed these as **standalone cases**.

Based on whether *cases* preserve or erode user privacy, they are pre-labeled with one of the following rating categories: *blocker*, *bad*, *neutral*, or *good*. Of the 130 cases, 63 were rated good, 35 bad, 25 neutral, and 7 blockers. Figure 6.1 shows a snapshot from ToS;DR for the service *YouTube* containing YouTube’s assigned letter grade, selected cases, and their respective ratings (Blocker: Red; Bad: Yellow; Good: Green; Neutral: Grey). Ratings indicate respect towards user privacy, where ‘blockers’ often indicate practices that are aggressively adversarial towards user privacy, while those rated ‘good’ indicate privacy-preserving practices. ‘Bad’ and ‘neutral’ fall in the middle.

6.2.1.2 Privacy Rating Analysis

We further investigated each ToS;DR privacy category, blocker, bad, good, and neutral by calculating the perplexity of the excerpts associated with them. Figure 6.2 shows the average perplexity of excerpts calculated using RoBERTa, a general purpose language model and PrivBERT, a language model pre-trained on privacy policy text. For comparison, we randomly sampled 1000 sentences from ten most visited websites’ policies. We calculated the perplexity by masking each word in the sequence separately and exponentiating the average the negative log likelihood of size 8 sequences with a stride of 4.

From Figure 6.2, it is evident that PrivBERT exhibits notably reduced perplexity scores in comparison to RoBERTa, a phenomenon attributed to PrivBERT’s pretraining on policy text. We hypothesize that for PrivBERT (but not RoBERTa), text segments commonly encountered in privacy policies should yield lower perplexity scores, while those that are infrequent should result in higher perplexity scores. Figure 6.2 shows that ‘blockers’ have the highest sentence perplexity scores for both language models suggesting that they might be written in particularly convoluted language unusual even within privacy policies. Conversely, other categories exhibit closely aligned perplexity

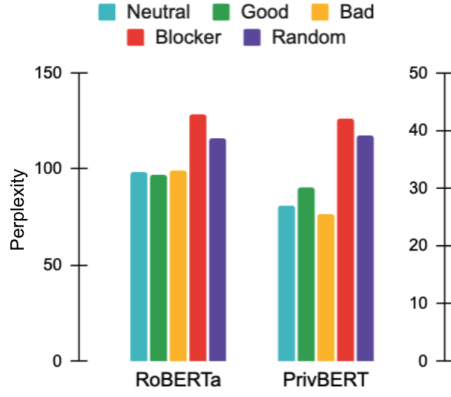


Figure 6.2: Perplexity distribution of ToS;DR ratings

scores for RoBERTa, but distinct variations for PrivBERT, suggesting that the pre-training data may encompass text associated with each category at varying frequencies. Notably, ‘bad’ practices are the second most perplexing category for RoBERTa, yet the least perplexing for PrivBERT, signifying the frequent inclusion of ‘bad’ practices-related text in policies. Conversely, this trend is reversed for ‘good’ practices, indicating their relative rarity.

6.2.1.3 Experiments

Model	C	Standalone Set				Contrasting Set			
		P	R	F1	S	P	R	F1	S
<i>RS</i>	0	0.94	0.95	0.94	2091	0.75	0.37	0.50	557
	1	0.95	0.94	0.94	2091	0.72	0.93	0.81	996
<i>CBS</i>	0	0.89	0.95	0.92	2091	0.71	0.82	0.76	557
	1	0.95	0.88	0.91	2091	0.95	0.91	0.93	996

Table 6.1: Performance of PrivBERT on random sampling and cluster-based sampling with a 1:3 sampling ratio (P: Precision, R: Recall, S: Support)

In this section, we describe our approach to automatically match excerpts from privacy policies to their respective ToS;DR *cases*. Formally, given an excerpt from a privacy policy, S_j , the task is to find the case(s) C_i that represent the data practice described in the excerpt S_j . Here, C is the set of 130 ToS;DR *cases*; S is the set of excerpts from a privacy policy. This problem can be framed as a multi-label classification task or a binary classification task. Preliminary experiments indicated that the multi-label classification was ineffective, due to a substantial imbalance in class distribution. Therefore, choose

a binary classification technique to train a model to distinguish between matching and disparate case-excerpt pairs.

Annotated case-excerpt pairs from ToS;DR serve as positive training samples. To collect negative samples, we use two sampling approaches, random sampling (RS) and cluster-based sampling (CBS).

Random Sampling (RS): For each annotated ToS;DR *case*-excerpt pair, we randomly sample excerpts matched with other *case* to serve as a negative sample irrespective of it being a *contrasting* or *standalone case*.

Cluster-Based Sampling (CBS): For each annotated *case*-excerpt pair, if the *case* is part of a cluster with contrasting *cases*, we first sample excerpts matched with other cases within the same cluster. We do this until we have either exhausted all possible unique negative samples¹ or we have the required number of negative samples. When the number of negative samples is not satisfied, we sample from outside the cluster randomly. For standalone cases, we collect negative samples randomly

We trained separate models for each sampling technique and sampling ratio. We experiment with sampling 1x, 2x, 3x, and 5x number of negative samples as positive samples. We divided the data into train, validation, and test sets in the ratio 3:1:1. We keep the test and validation sets constant across different sampling ratios, with 1:1 sampling. To train the model, we input the concatenated case C_i and segment S_j separated by a special token to fine-tune PrivBERT, a privacy policy language model (Srinath, Wilson, & Giles, 2021). We use a binary cross-entropy loss function to optimize the parameters and train the model using Adam optimizer (Kingma & Ba, 2014) with a learning rate of 1e-5 for 3 epochs.

We evaluate the trained model on test sets each containing policy excerpts previously unseen by the model. The standalone set and the contrasting set contain case-excerpt pairs from standalone cases contrasting cases respectively. We compare the performance of our model to several approaches listed in Table 6.2. 1) We use Sentence-BERT (Reimers & Gurevych, 2019) to vectorize policy excerpts and ToS;DR cases. We then calculate the cosine distance between each pair and identify a threshold similarity score (0.25) based on the validation set. 2) We prompt OpenAI’s GPT-4 to identify matching case-excerpt pairs 3) We enhance the GPT-4 prompt with two examples, one positive/negative case-excerpt pair 4) We train RoBERTa (Y. Liu et al., 2019) recreating the experiments with PrivBERT and report the results on the best performing model.

¹This can happen when only one of the *cases* in a cluster has a large number of positive samples

6.2.1.4 Loss Function

Given a *case* from ToS;DR, C_i , and an excerpt from a privacy policy, S_j , where,

$C \in \text{set of ToS;DR cases};$

$S \in \text{set of privacy policy excerpts}$

The loss function is given by,

$$-y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}) \quad (6.1)$$

$$y = \text{Match}(C_i, S_j) \quad (6.2)$$

6.2.2 Results

The results of PrivBERT trained on datasets created by random sampling (RS) and cluster-based sampling (CBS), for a 1:3 sampling ratio, is shown in Table 6.1. The results in the table are reported after taking the mean (max standard deviation: 0.05) on 3 runs with different random initial states and training sets shuffled. The 0 category refers to disparate case-excerpt pairs while 1 refers to matching pairs. The table shows that the results on the standalone set are superior to those on the contrasting case set across both sampling techniques. This is likely since *contrasting cases* are often lexically and semantically quite similar, and therefore a harder problem. We also see that cluster-based sampling performs significantly better than random sampling on the contrasting set while maintaining a similar performance on the standalone set. This is expected since a larger number of samples are used to train the model to solve the harder problem of disambiguating between similar cases.

For the RS model, the results on the contrasting set improved with larger negative sampling ratios while the CBS model remained the same. This could indicate that a larger number of contrasting case training samples were necessary for the RS model. On the other hand, the results for both the models on the standalone set remained constant across different sampling ratios. Results for different sampling ratios are presented in the appendix.

We achieve state-of-the-art results on this dataset, the performance of several other approaches is shown in Table 6.2. Finetuned RoBERTa-base is the second best-performing model, followed by the two-shot version of GPT-4. The performance improvement due

Model	C	P	R	F1
Sentence-BERT	0	0.61	0.66	0.63
	1	0.78	0.71	0.74
GPT-4 (Zero-shot)	0	0.72	0.74	0.73
	1	0.82	0.80	0.81
GPT-4 (2-shot)	0	0.79	0.74	0.76
	1	0.83	0.86	0.84
RoBERTa (CS)	0	0.78	0.85	0.81
	1	0.91	0.87	0.89
PrivBERT (CS)	0	0.80	0.88	0.83
	1	0.95	0.89	0.92

Table 6.2: Test performance (P: Precision, R: Recall)

to fine-tuning depicts the complexity of the task. Sentence-BERT performs well on case-excerpt pairs with significant lexical overlap.

Under real-world conditions, identifying all data practices within a privacy policy would involve testing whether an excerpt matches any case in the full set of ToS;DR cases. It is therefore important that the model be trained robustly on dissimilar pairs, supporting training on larger sampling ratios and providing high precision scores for negatives. Furthermore, the erroneous classification of a pair as a ‘match’ can be readily discerned and flagged, thus facilitating the establishment of a feedback loop. Conversely, the misclassification of a pair as dissimilar may evade detection, reinforcing our argument for the necessity of robustly performing models on dissimilar pairs.

6.2.2.1 Case Study

We present a case study evaluating our model’s performance on a previously un-annotated policy. We chose Airbnb², a popular online marketplace, due to its popularity and its legitimate need to access sensitive personal user information. We split the policy based on new line characters, resulting in 150 segments encompassing paragraphs, section titles, and list items. Two authors of the paper thoroughly reviewed each segment and tagged them with any associated ToS;DR *cases*. Subsequently, we employed the best-performing CBS model to identify matching *case-excerpt* pairs. Table 6.3 that the model consistently identified true positives, while producing a few false positives, therefore leading to a high recall for matching case-excerpt pairs. Most false positives were caused due to inaccurate predictions on privacy policy section titles. The model was able to identify

²<https://www.airbnb.com/>

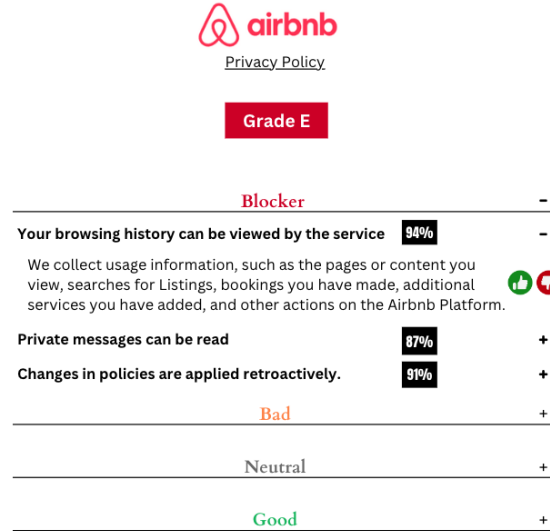


Figure 6.3: Privacy label with varying information levels. Detailed information access by the expand/contract feature. Model probabilities shown as %; thumbs-up/down icons for user feedback on matches.

true negatives with a high degree of accuracy while producing a few false negatives, leading to high precision scores for dissimilar case-excerpt pairs.

	Precision	Recall	F1	Support
True	0.79	0.88	0.83	285
False	0.87	0.84	0.85	19,215

Table 6.3: Case-level results

In Fig. 6.3, we demonstrate the tool that provides users with privacy information at varying levels of granularity. At the highest level, identified *cases* are summarized using a letter grade with the grading scheme adopted by ToS;DR: 1 or more ‘blockers’: Grade E — 75% ‘bad’ *cases*: Grade D — 50% - 75% ‘bad’ *cases*: Grade C — 50% ‘bad’ *cases*: Grade B — 25% ‘bad’ *cases*: Grade A. At the next level, identified *cases* are listed under their rating categories. Finally, at the most fine-grained level policy excerpts associated with a case are listed along with a match probability score.

6.2.3 Conclusion

Our research introduces a novel solution to the challenge of simplifying complex privacy policies. We provide an automated approach to scale identification of data practices

in privacy policies, providing users with information regarding their data privacy. By automating the process of aligning data practices with policy excerpts, we are taking a crucial step towards eliminating the need for users to laboriously parse through lengthy and intricate documents, ensuring that they receive precise and easily digestible information about their online privacy. This lowers the barrier for internet users to understand what is happening with their data online, thereby allowing them to make informed privacy decisions. While there is room for further refinements and research, our model presents a promising foundation for future advancements in the field of usable privacy, ultimately promoting greater user empowerment and security in the digital age.

6.2.4 Prompts Used for GPT-4

We used the OpenAI API GPT-4 models with a temperature setting of 0.

You are a privacy policy expert. Given a title and quote, your task is to evaluate whether the title represents the data practice described in the quote. Your output should only be either 0 (indicating the title does not represent the quote) or 1 (indicating the title represents the quote).

Example 1

Title: Third parties are involved in operating the service.

Quote: Note that we don't use any 3rd party website statistics tools like Google Analytics or similar.

Output: 1

Example 2

Title: You can opt out of targeted advertising

Quote: Our CDN is Cloudflare, and they may include cookies with our pages to provide a better service.

Output: 0

Title: {title}

Quote: {quote}

6.3 Automatic Date Extraction

Extracting dates from policies is crucial for informing trends in the online privacy landscape and help study historical user privacy cause-and-effect relationships. Systematically extracting dates at scale from policies will aid in privacy regulation enforcement. The GDPR (Articles 12-14), explicitly requires transparent disclosure of information, accurately reflecting data practices. The lack of a policy date would create doubt about whether the policy is reflecting up-to-date data practices therefore violating this requirement in principle. The privacy policy is the only document that consumers can use to understand what happens with their personal information, and without a policy date, a consumer would not be able to ascertain if a policy is current or up-to-date. Additionally, the lack of posted policy dates hampers regulatory oversight, by making it difficult or impossible to determine when a policy was effective. Here, we investigate whether the policy is up-to-date, and study the effects of laws and regulations on policy updates.

We use four large web crawls discussed above, containing over 3.5 million privacy policies, to study trends in dates and updates in privacy policies online. We create a pipeline to extract policy dates by first extracting text from the privacy policy HTML, followed by extracting all date instances and classifying candidate policy dates into either policy date, i.e., *updated date* or *effective date*, or other types of dates using a transformer-based language model³. We find that less than 40% of policies online have a policy date and that popular domains are more likely to have a policy date. Finally, we investigate trends in updating privacy policies and find that the dates when new privacy laws became effective have had a significant impact on when most privacy policies have been updated. Our findings strongly suggest that a large percentage of policies online are not up-to-date and are not compliant with legal requirements.

6.3.1 Related Work

Research on extracting dates from unstructured documents has focused on the news domain, and used named entity recognition (NER) approaches. TimeLineCurator extracts all dates in news articles using a supervised NER approach (Fulda et al., 2015). News articles are often dense in dates since they recount events. NER is thus the most appropriate technique since temporal features could be recounted in various ways, such as the day of the week, day of the month, etc. Similarly, Smith (2002) finds date-place

³Data and code available at <https://privaseer.ist.psu.edu/data>

co-locations in news articles.

Work in unstructured date extraction, similar to that in this paper, can be found in the medical domain. Fu et al. (2020) extract diagnosis dates from clinical notes using a regex approach followed by a machine learning model to classify the extracted dates. However, they test their approach on a very small document set with dates occurring in similar formats. Similarly, studies on extracting clinical information from patient health records often also include extracting a date (Shah et al., 2012; Fonferko-Shadrach et al., 2019). However, they often contain much richer context, thereby helping to classify date types.

Prior research studying patterns in privacy policies’ dates is scarce. Linden et al. (2020) and Degeling et al. (2018) studied how GDPR changed the privacy landscape using the Wayback Machine to download policies before and after the GDPR and compared changes between them. PrivaSeer, the privacy policy search engine (Srinath, Sundareswara, et al., 2021) displays the date that a policy was previously crawled and makes them searchable. However, no prior work concentrates on extracting dates from policies in order to analyze policy update patterns.

6.3.2 Policy Date Extraction

We attempt to solve the problem of extracting the updated and effective dates given a large corpus of privacy policies while optimizing for time and computational resources. We create a policy date extraction pipeline (Srinath, Wilson, & Giles, 2021; Sundareswara et al., 2021) in which we first extract text snippets containing candidate policy dates using a regex-based approach. We then classify the candidate instances and construct date objects using an open-source date extraction tool. We then analyze trends in dates in over 3.5 million web privacy policies using four large-scale web crawls between May 2019 and September 2021.

Regex Extraction: We surveyed a number of tools that automatically extract dates from unstructured text. Table 6.4 shows the tools and how they perform on a random sample of one hundred documents. The tools surveyed, namely, Spacy (Honnibal et al., 2020), SUTtime (Chang & Manning, 2012), dateparser⁴ and datefinder⁵ extract too many non-policy date instances with dateparser and datefinder extracting almost 22 instances per policy, thereby requiring more processing further down the pipeline. SUTime takes almost 5s per policy on a single CPU core, translating to about half a

⁴<https://github.com/scrapinghub/dateparser>

⁵<https://github.com/akoumjian/datefinder>

year to extract date instances from 3.5 million policies. Although Spacy’s named entity recognition (NER) approach achieves reasonable results, our custom approach achieves a multi-fold improvement on time and precision. Our custom regex concentrates on matching common date patterns in privacy policies, such as, four-digit numbers starting with 20 or 19, (any policy would need to be written between the late 1990s and today) and a combination of separators, date, month and year. We then extract at most 250 characters (or up to a newline character) of text before and after the regex match to capture the context of the matched four-digit number.

Tool	Time (s)	Instances	P	R
datefinder	0.0005	21.8	0.01	0.75
dateparser	0.1	21.8	0.01	1
SUTime	5	5.5	0.04	1
Spacy (NER)	0.24	4.6	0.05	1
Custom Regex	0.0005	3.0	0.13	1

Table 6.4: Performance of open-source date extraction tools (P: Precision; R: Recall; time (seconds) indicated per policy; instances (mean) per policy)

Date Instance Classification: Privacy policies contain a number of date instances. Most commonly, they contain when the policy was updated, when the policy went into effect, date of previous revision, version approval date, and regulation enforcement date. Since we were interested in studying the trends of updating privacy policies, we concentrated on extracting *updated* and *effective* dates. We refer to these as ‘policy dates’. The following are a few examples of types of policy date instances we extract: *Effective: January 1st, 2022, This policy went into effect on 1/1/22, The policy dated January 1, 2022, replaces all previous policies, Last updated: January 2022, Last Revised 01/01/2022, This page was last edited on 1 Jan 2022.*

We use a supervised classification approach to separate policy dates from all the date instances extracted. One of the authors, a privacy policy expert, labeled 1,075 uniformly randomly sampled date instances. 230 were found to be *updated dates*, 170 were *effective dates* and the rest (675) were *other* date instances. We trained two models to classify the extracted date instances. We trained a random-forest model with 100 estimators and a minimum of 2 samples to split a node on tf-idf features. We also fine-tuned PrivBERT (Srinath, Wilson, & Giles, 2021), a language model pretrained on privacy policies, using the Adam optimizer with a training rate of 1e-4 and batch size of 64 for three epochs. We divided the labeled documents into train, test, and validation sets in the ratio 3:1:1. Table 3.1 shows the results for this task. We can see that PrivBERT improves the

performance of the baseline approach by a few percentage points.

Class	Random Forest			PrivBERT			S
	P	R	F1	P	R	F1	
Updated	0.98	0.92	0.95	0.95	0.97	0.96	48
Effective	0.95	0.92	0.93	0.97	0.97	0.97	38
Other	0.97	1.0	0.98	0.99	0.99	0.99	129

Table 6.5: Date instance classification results (S: Support)

Date Object Extraction: After the date instances were classified as either a policy date or not, we used dateparser to create date objects, i.e., separately representing date text instances into day, month, and year. A number of tools exist that convert date instances into objects, some of which we surveyed in the section on regex extraction shown in Table 6.4. We used dateparser since it was quick, had a high accuracy rate, and was the most convenient.

6.3.3 Policy Date Analysis

We created a pipeline to efficiently collect privacy policies at scale and used four large-scale web crawls to study date patterns in them. Our pipeline involved first collecting candidate privacy policy URLs from the CommomCrawl URL dump using a selection criterion: URLs containing the words ‘privacy’ or the words ‘data’ and ‘protection.’ We then crawled the URLs satisfying this criterion and filtered the English language documents using LangID (Lui & Baldwin, 2012a). Next, we trained a model to classify whether a given document is a privacy policy (Srinath, Wilson, & Giles, 2021). We obtained a precision of 0.98, and a recall of 0.98 (Srinath, Wilson, & Giles, 2021). Next, we removed duplicates by hashing the HTML pages, and removed near-duplicates by creating simhashes (similar items have similar hashes) (Charikar, 2002) and excluding documents based on a Hamming distance threshold (Manku et al., 2007).

We undertook the first crawl in May 2019 and obtained about 1 million privacy policies (Srinath, Wilson, & Giles, 2021). We completed the second crawl in March 2020, starting with a LinkedIn company dataset, and obtained about 600k policies (Sundareswara et al., 2020). We completed the third crawl in March 2021, starting with a more recent CommonCrawl archive, and created a collection of about 1.2 million policies. Finally, we undertook the fourth crawl in September 2021, when we re-crawled all the URLs in the first, and second crawls. Overall, we collected over 3.5 million policies, including multiple versions of the same policy.

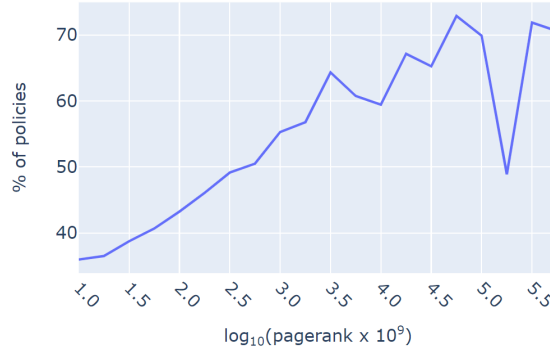


Figure 6.4: Relationship between %policies containing at least one date instance and the domain pagerank value

We put all four crawls through the date extraction pipeline. We found that only a small portion of the policies in any crawl had a policy date, with the March 2020 crawl containing the highest percentage (41.6%) compared to 36.6% and 38% in the May 2019 and March 2021 crawls respectively. This is likely due to the source of the seed URLs used. The seed URLs for the March 2019 crawl came from the company URLs listed on LinkedIn, whereas CommonCrawl was the source used for the other crawls. CommonCrawl seeds are more likely to contain links to hobby sites, information sites, and other non-commercial sites that are less likely to maintain up-to-date privacy policies and follow regulatory updates. More evidence for this hypothesis can be found in Figure 6.4, which shows how the presence of policy dates varies based on the pagerank of the domain as calculated from the CommonCrawl web-graph. In the figure, the x-axis contains bins of pagerank values, where the higher the pagerank value, the more popular the domain. Each bin contains at least 50 policies, with the number of policies skewed heavily towards the left (less popular domains). We can see that more popular domains are more likely to contain policy date instances. This is likely due to the fact that more popular sites face heavier regulatory scrutiny.

Figure 6.5 shows that a large percentage of policies have a policy date in May 2018. This corresponded with when GDPR came into effect, i.e., May 25, 2018. Degeling et al. (2018) noted a similar increase in updates to privacy policies leading up to and around the GDPR effective date. This shows that many policies updated to comply with GDPR have not been updated in the four years since. Further, the second most significant spike in the graph occurred in December 2019, corresponding with the CCPA coming into effect starting 1 January 2020. The earlier peak around April 2003, corresponds with the passage of the Privacy Act of 2003. These peaks in the policy update patterns surround-

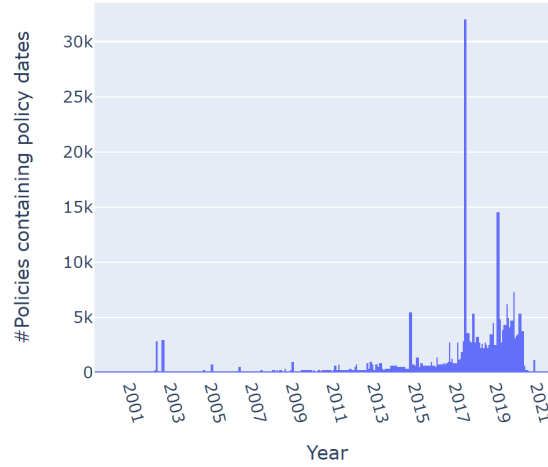


Figure 6.5: Policy date distribution

ing regulation enforcement dates suggest that organizations updated their policies with the passage of new regulations but failed to keep their policies regularly updated.

We investigated how often policies were updated by measuring the number of overlapping policies between each crawl and then counting the number of policies whose policy dates changed. Between the May 2019 and September 2021 crawls, i.e., in over two years, we found that only about 32% of policies were updated. Between the May 2019 and March 2020 crawls, 27% of policies were updated, while between the March 2020 and March 2021 crawls, 29% of policies were updated, respectively.

Finally, we investigated the likelihood of policies having a date based on their sector of commerce. (Sundareswara et al., 2020) categorized the industry information (obtained through LinkedIn) for all the domains in the March 2020 crawl into ten sectors of commerce. We found that the *Information Technology* sector and the *Education* sector has the highest percentage of policies, with 42.24% and 42.1% of policies having a policy date respectively, while *Consumer and supply chain* sector is the one with the lowest percentage of policies, with around 39.7%, containing a policy date. Overall, we hypothesize that sectors that are more likely to be consumer-facing have a higher likelihood of having policy dates.

6.3.4 Discussion

The privacy policy is the only document consumers can use to understand what happens with their personal information online. It is, therefore, paramount that organizations reflect up-to-date information regarding their data practices on their policies. Figure 6.5

shows how a number of policies have been dormant for up to 20 years (2000-2020). In contrast, real-world data collection practices have undergone a multitude of significant changes during the same period, suggesting that these policies may not reflect up-to-date data collection practices of their respective organizations. A lack of policy dates can thus create confusion regarding whether a policy is being updated or maintained from a user’s point of view and thus disincentivizing them to read policies, and further contributing to the failures of the notice and choice model. Our findings shown in Figure 6.4 suggest that even among popular domains (with a high pagerank), only about 70% contain a policy date with fewer updating their policies annually. This percentage falls to less than 40% of policies when considering less popular domains. Additionally, our findings show that at scale, this trend has held steady between May 2019 and September 2021 suggesting that a more stringent regulatory incentive is required for organisations to include a policy dates in their documents and keep them updated.

We created a novel pipeline to extract updated and effective dates from privacy policies at scale while optimizing for time and computational resources. The pipeline includes crawling, regex extraction, date instance classification, and date object extraction, and can be applied to extract date instances at scale for any domain by retraining the machine learning model. We crawled and applied the pipeline on over 3.5 million privacy policies and extracted policy dates from them. We found that less than 40% of policies have at least a single policy date and less than 30% of which are updated annually. Further, we found that policies tend to be updated en mass as a consequence of new regulation enforcement and that more popular domains are more likely to update policies or even contain a policy date. Finally, we found that the consumer-facing sectors had the highest percentage of policies with dates, with about 42% containing a policy date.

6.4 Conclusion

In conclusion, the complexity of privacy policies poses a significant challenge for users wishing to understand how their data is handled. This chapter underscores the necessity of simplifying privacy information to empower users in making informed decisions about their online privacy. Existing methods, such as privacy nutrition labels, have shown potential but are not yet widely adopted or fully effective.

Our approach focuses on the automatic extraction of key information, specifically data practices and dates, from privacy policies. By leveraging natural language pro-

cessing (NLP) techniques, we aim to provide users with clear and concise summaries of privacy practices. This automated solution seeks to reduce the burden on users, making it easier for them to understand and manage their online privacy. Ultimately, enhancing the accessibility of crucial privacy information through scalable and effective NLP techniques can lead to a more user-friendly and privacy-conscious online environment.

Chapter 7 —

Conclusion and Future Work

This thesis explored the challenges associated with online user privacy and the limitations of users in understanding privacy policies. We argued that the prevalence of lengthy and legally-oriented privacy policies hinders user comprehension and informed consent. To address this issue, we proposed a multi-faceted approach that leverages natural language processing (NLP) techniques.

Our first contribution was the creation of a large-scale privacy policy corpus, *PrivaSeer*. This corpus serves as a valuable resource for researchers and developers working on privacy-related NLP tasks. We then developed *PrivBERT*, a fine-tuned language model specifically designed for the privacy policy domain. *PrivBERT* demonstrated improved performance on question answering and text classification tasks compared to generic language models. Next, we investigated the availability of privacy policies online. We identified significant challenges, including dead links, non-standard locations for privacy policies, and the presence of non-policy documents disguised as privacy policies. Based on this analysis, we estimated a lower bound for the total number of privacy policies on the internet. We then presented a privacy policy search engine that allows users to search for specific data practices across different policies. This search engine leverages *PrivBERT* for indexing and ranking, enabling users to compare privacy practices of different organizations. Finally, we explored approaches to simplify privacy policies to empower users with better control over their data. We developed a system for automatically extracting data practices from privacy policies. Additionally, we explored techniques for automatically extracting the date a privacy policy was last updated.

Overall, this work demonstrates the potential of NLP for improving the online privacy landscape. By making privacy policies more accessible and understandable, we can empower users to make informed decisions about their data and hold organizations accountable for their data practices.

The cumulative contributions of this thesis significantly advance our understanding of online privacy in several crucial ways. First, by identifying and addressing the fundamental challenges in the accessibility of privacy policies, we highlight the necessity of reforming how organizations communicate their data practices. This shift not only augments user comprehension but also facilitates a more informed public discourse surrounding privacy rights.

Furthermore, the creation of the PrivaSeer corpus and the PrivBERT model offers new resources and methods for future research. These tools are likely to encourage deeper exploration of privacy policies, potentially leading to a better understanding of user behaviours and attitudes toward privacy. This work sets itself apart from existing literature by using advanced natural language processing techniques to analyze privacy policies, filling an important gap in current research.

In addition, the introduction of the privacy policy search engine represents a transformative step in regulatory engagement with privacy information. By enabling regulators to search and compare data practices across various organizations, this tool addresses the pressing need for resources that enhance oversight and accountability in data management. Such capabilities not only facilitate more informed regulatory decisions but also promote transparency in organizational practices, ultimately fostering a culture of compliance and trust in the digital landscape.

In summary, this thesis enhances our understanding of online privacy challenges and establishes a foundation for future research and practical applications that can help create a more informed and empowered internet user community. The implications of our findings are relevant to internet users, policymakers, organizations, and individuals who seek a more transparent and fair digital environment.

7.1 Future Work

Our work opens several exciting avenues for future research. Here we discuss a few potential directions:

1. **Improving Privacy-Enhancing Technologies (PETs):** The accuracy and reliability of PETs used for extracting data practices and recognizing privacy policy updates can be refined. Additionally, there is potential to develop PETs that summarize key points from privacy policies or visually represent data flows, thereby simplifying complex information for users.

2. **Incorporating User Feedback:** The effectiveness of privacy policy search engines and PETs can be enhanced through the integration of user feedback mechanisms. This would allow the system to learn from user interactions, thereby improving its capabilities and better-addressing user needs and preferences.
3. **Multilingual Support:** Although English is the most popular language for privacy policies, a significant number is in other languages. Expanding research efforts to include multiple languages would significantly broaden the reach and impact of privacy policy tools, making essential information more accessible to a global audience.
4. **Regulatory Compliance Automation:** Natural language processing (NLP) techniques can be utilized to automate tasks related to privacy policy compliance for organizations. This could involve creating tools that identify potential violations of privacy policies and recommend necessary improvements to ensure regulatory adherence.
5. **Longitudinal Studies:** Conducting longitudinal studies of privacy policies can provide valuable insights into the evolution of data practices and the influence of regulations over time. Such studies can inform both academic research and practical policy-making.

References

- Aberkane, A.-J., Broucke, S. V., & Poels, G. (2022). Investigating organizational factors associated with gdpr noncompliance using privacy policies: A machine learning approach. In *2022 ieee 4th international conference on trust, privacy and security in intelligent systems, and applications (tps-isa)* (pp. 107–113).
- Ackerman, M. S., Cranor, L. F., & Reagle, J. (1999). Privacy in e-commerce: examining user scenarios and privacy preferences. In *Proceedings of the 1st acm conference on electronic commerce* (pp. 1–8).
- Amos, R., Acar, G., Lucherini, E., Kshirsagar, M., Narayanan, A., & Mayer, J. (2020). Privacy policies over time: Curation and analysis of a million-document dataset. *arXiv preprint arXiv:2008.09159*.
- Arasu, A., Novak, J., Tomkins, A., & Tomlin, J. (2002). Pagerank computation and the structure of the web: Experiments and algorithms. In *Proceedings of the eleventh international world wide web conference, poster track* (pp. 107–117).
- Assembly, C. S. (2020). *California consumer privacy act*. https://leginfo.ca.gov/faces/codes_displayText.xhtml?division=3&part=4.&lawCode=CIV&title=1.81.5.
- Bannihatti Kumar, V., Iyengar, R., Nisal, N., Feng, Y., Habib, H., Story, P., ... others (2020). Finding a choice in a haystack: Automatic extraction of opt-out statements from privacy policy text. In *Proceedings of the web conference 2020* (pp. 1943–1954).
- Barnes, S. B. (2006). A privacy paradox: Social networking in the united states. *First Monday*, 11(9).
- Barth, S., & De Jong, M. D. (2017). The privacy paradox—investigating discrepancies between expressed privacy concerns and actual online behavior—a systematic literature review. *Telematics and informatics*, 34(7), 1038–1058.
- Baykan, E., Henzinger, M., Marian, L., & Weber, I. (2009). Purely url-based topic classification. In *Proceedings of the 18th international conference on world wide web* (pp. 1109–1110).

- Begon, M., et al. (1979). *Investigating animal abundance: capture-recapture for biologists*. Edward Arnold (Publishers) Ltd.
- Beltagy, I., Cohan, A., & Lo, K. (2019). Scibert: Pretrained contextualized embeddings for scientific text. *arXiv preprint arXiv:1903.10676*.
- Bhatia, J., Breaux, T. D., Reidenberg, J. R., & Norton, T. B. (2016). A theory of vagueness and privacy risk perception. In *2016 IEEE 24th International Requirements Engineering Conference (re)* (pp. 26–35).
- Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. *Journal of machine Learning research*, 3(Jan), 993–1022.
- Böhning, D., Rocchetti, I., Maruotti, A., & Holling, H. (2020). Estimating the undetected infections in the covid-19 outbreak by harnessing capture–recapture methods. *International Journal of Infectious Diseases*, 97, 197–201.
- Brenner, H. (1995). Use and limitations of the capture-recapture method in disease monitoring with two dependent sources. *Epidemiology*, 42–48.
- Broder, A. Z., Glassman, S. C., Manasse, M. S., & Zweig, G. (1997). Syntactic clustering of the web. *Computer networks and ISDN systems*, 29(8-13), 1157–1166.
- Byers, S., Cranor, L. F., Kormann, D., & McDaniel, P. (2004). Searching for privacy: Design and implementation of a p3p-enabled search engine. In *International workshop on privacy enhancing technologies* (pp. 314–328).
- Chakrabarti, S., Van den Berg, M., & Dom, B. (1999). Focused crawling: a new approach to topic-specific web resource discovery. *Computer networks*, 31(11-16), 1623–1640.
- Chalkidis, I., Fergadiotis, M., Malakasiotis, P., Aletras, N., & Androutsopoulos, I. (2020, November). LEGAL-BERT: The muppets straight out of law school. In T. Cohn, Y. He, & Y. Liu (Eds.), *Findings of the association for computational linguistics: Emnlp 2020* (pp. 2898–2904). Online: Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2020.findings-emnlp.261> doi: 10.18653/v1/2020.findings-emnlp.261
- Chalkidis, I., Jana, A., Hartung, D., Bommarito, M., Androutsopoulos, I., Katz, D., & Aletras, N. (2022, May). LexGLUE: A benchmark dataset for legal language understanding in English. In S. Muresan, P. Nakov, & A. Villavicencio (Eds.), *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 4310–4330). Dublin, Ireland: Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2022.acl-long.297> doi: 10.18653/v1/2022.acl-long.297
- Chang, A. X., & Manning, C. D. (2012). SUTIME: A library for recognizing and normalizing time expressions. In *Lrec* (Vol. 3735, p. 3740).

- Charikar, M. S. (2002). Similarity estimation techniques from rounding algorithms. In *Proceedings of the thirty-fourth annual acm symposium on theory of computing* (pp. 380–388).
- Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). Smote: synthetic minority over-sampling technique. *Journal of artificial intelligence research*, 16, 321–357.
- Costante, E., Sun, Y., Petković, M., & den Hartog, J. (2012). A machine learning solution to assess privacy policy completeness: (short paper). In *Proceedings of the 2012 acm workshop on privacy in the electronic society* (pp. 91–96).
- Cranor, L. (2002). *Web privacy with p3p*. ” O’Reilly Media, Inc.”.
- Cranor, L. F. (2012). Necessary but not sufficient: Standardized mechanisms for privacy notice and choice. *J. on Telecomm. & High Tech. L.*, 10, 273.
- Cranor, L. F., Arjula, M., & Guduru, P. (2002). Use of a p3p user agent by early adopters. In *Proceedings of the 2002 acm workshop on privacy in the electronic society* (pp. 1–10).
- Cranor, L. F., Hoke, C., Leon, P. G., & Au, A. P. (2014). Are they worth reading? an in-depth analysis of online advertising companies’ privacy policies..
- Cranor, L. F., Idouchi, K., Leon, P. G., Sleeper, M., & Ur, B. (2013). Are they actually any different? comparing thousands of financial institutions’ privacy practices. In *Proc. weis* (Vol. 13).
- Davis, M., & Iancu, L. (2012). Unicode text segmentation. *Unicode Standard Annex*, 29.
- Degeling, M., Utz, C., Lentzsch, C., Hosseini, H., Schaub, F., & Holz, T. (2018). We value your privacy... now take some cookies: Measuring the gdpr’s impact on web privacy. *arXiv preprint arXiv:1808.05096*.
- Deuker, A. (2009). Addressing the privacy paradox by expanded privacy awareness—the example of context-aware services. In *Ifip primelife international summer school on privacy and identity management for life* (pp. 275–283).
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2018). Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the north american chapter of the association for computational linguistics: Human language technologies, volume 1 (long and short papers)* (pp. 4171–4186).

- Diligenti, M., Coetzee, F., Lawrence, S., Giles, C. L., & Gori, M. (2000). Focused crawling using context graphs. In *Vldb*.
- Dobra, A., & Fienberg, S. E. (2004). How large is the world wide web? In *Web dynamics* (pp. 23–43). Springer.
- Earp, J. B., Antón, A. I., Aiman-Smith, L., & Stufflebeam, W. H. (2005). Examining internet privacy policies within the context of user privacy values. *IEEE Transactions on Engineering Management*, 52(2), 227–237.
- Earp, J. B., & Baumer, D. (2003). Innovative web use to learn about consumer behavior and online privacy. *Communications of the ACM*, 46(4), 81–83.
- Ermakova, T., Fabian, B., & Babina, E. (2015). Readability of privacy policies of healthcare websites. *Wirtschaftsinformatik*, 15.
- Ester, M., Kriegel, H.-P., Sander, J., Xu, X., et al. (n.d.). A density-based algorithm for discovering clusters in large spatial databases with noise..
- Fabian, B., Ermakova, T., & Lentz, T. (2017). Large-scale readability analysis of privacy policies. In *Proceedings of the international conference on web intelligence* (pp. 18–25).
- Factsheet 5 - what to expect when we inspect*. (2018). Retrieved 2020-05-11, from https://edps.europa.eu/sites/edp/files/publication/factsheet_5_en.pdf
- Flender, C., & Müller, G. (2012). Type indeterminacy in privacy decisions: the privacy paradox revisited. In *International symposium on quantum interaction* (pp. 148–159).
- Fonferko-Shadrach, B., Lacey, A. S., Roberts, A., Akbari, A., Thompson, S., Ford, D. V., ... Pickrell, W. O. (2019). Using natural language processing to extract structured epilepsy data from unstructured clinic letters: development and validation of the exect (extraction of epilepsy clinical text) system. *BMJ open*, 9(4), e023232.
- Fu, J. T., Sholle, E., Krichevsky, S., Scandura, J., & Champion, T. R. (2020). Extracting and classifying diagnosis dates from clinical notes: a case study. *Journal of Biomedical Informatics*, 110, 103569.
- Fulda, J., Brehmer, M., & Munzner, T. (2015). Timelinecurator: Interactive authoring of visual timelines from unstructured text. *IEEE transactions on visualization and computer graphics*, 22(1), 300–309.
- Gormley, C., & Tong, Z. (2015). *Elasticsearch: the definitive guide: a distributed real-time search and analytics engine*. ” O’Reilly Media, Inc.”.
- Gottron, T. (2007). Evaluating content extraction on html documents. In *Proceedings of the 2nd international conference on internet technologies and applications (ita’07)* (pp. 123–132).

- Gururangan, S., Marasović, A., Swayamdipta, S., Lo, K., Beltagy, I., Downey, D., & Smith, N. A. (2020). Don't stop pretraining: Adapt language models to domains and tasks. *arXiv preprint arXiv:2004.10964*.
- Harkous, H., Fawaz, K., Lebre, R., Schaub, F., Shin, K. G., & Aberer, K. (2018). Polisis: Automated analysis and presentation of privacy policies using deep learning. In *27th {USENIX} security symposium ({USENIX} security 18)* (pp. 531–548).
- Hoke, C., Cranor, L. F., Leon, P. G., & Au, A. P. (2015). Are they worth reading? an in-depth analysis of online trackers' privacy policies. *I/S: A Journal of Law and Policy for the Information Society*.
- Honnibal, M., Montani, I., Van Landeghem, S., & Boyd, A. (2020). spaCy: Industrial-strength Natural Language Processing in Python.
- Jensen, C., & Potts, C. (2004). Privacy policies as decision-making tools: an evaluation of online privacy notices. In *Proceedings of the sigchi conference on human factors in computing systems* (pp. 471–478).
- Kalamkar, P., Agarwal, A., Tiwari, A., Gupta, S., Karn, S., & Raghavan, V. (2022, December). Named entity recognition in Indian court judgments. In N. Aletras, I. Chalkidis, L. Barrett, C. Goanță, & D. Preotiuc-Pietro (Eds.), *Proceedings of the natural legal language processing workshop 2022* (pp. 184–193). Abu Dhabi, United Arab Emirates (Hybrid): Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2022.nllp-1.15> doi: 10.18653/v1/2022.nllp-1.15
- Kelley, P. G., Bresee, J., Cranor, L. F., & Reeder, R. W. (2009). A" nutrition label" for privacy. In *Proceedings of the 5th symposium on usable privacy and security* (pp. 1–12).
- Kelley, P. G., Cesca, L., Bresee, J., & Cranor, L. F. (2010). Standardizing privacy notices: an online study of the nutrition label approach. In *Proceedings of the sigchi conference on human factors in computing systems* (pp. 1573–1582).
- Kelley, P. G., Cranor, L. F., & Sadeh, N. (2013). Privacy as part of the app decision-making process. In *Proceedings of the sigchi conference on human factors in computing systems* (pp. 3393–3402).
- Khabsa, M., & Giles, C. L. (2014). The number of scholarly documents on the public web. *PloS one*, 9(5), e93949.
- Kincaid, J. P., Fishburne Jr, R. P., Rogers, R. L., & Chissom, B. S. (1975). Derivation of new readability formulas (automated readability index, fog count and flesch reading ease formula) for navy enlisted personnel.
- Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*.

- Klare, G. R., et al. (1963). Measurement of readability.
- Kohlschütter, C., Fankhauser, P., & Nejdl, W. (2010). Boilerplate detection using shallow text features. In *Proceedings of the third acm international conference on web search and data mining* (pp. 441–450).
- Kretschmer, M., Pennekamp, J., & Wehrle, K. (2021). Cookie banners and privacy policies: Measuring the impact of the gdpr on the web. *ACM Transactions on the Web (TWEB)*, 15(4), 1–42.
- Kumar, P. C. (2016). Privacy policies and their lack of clear disclosure regarding the life cycle of user information. In *Aaai fall symposia*.
- Kumar, V. B., Ravichander, A., Story, P., & Sadeh, N. (2019). Quantifying the effect of in-domain distributed word representations: A study of privacy policies. In *Aaai spring symposium on privacy-enhancing artificial intelligence and language technologies*.
- Lawrence, S., & Giles, C. L. (1998). Searching the world wide web. *Science*, 280(5360), 98–100.
- Le, Q., & Mikolov, T. (2014). Distributed representations of sentences and documents. In *International conference on machine learning* (pp. 1188–1196).
- Lebanoff, L., & Liu, F. (2018). Automatic detection of vague words and sentences in privacy policies. In *Proceedings of the 2018 conference on empirical methods in natural language processing* (pp. 3508–3517).
- Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J. (2020). Biobert: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4), 1234–1240.
- Leitner, E., Rehm, G., & Moreno-Schneider, J. (2020, May). A dataset of German legal documents for named entity recognition. In N. Calzolari et al. (Eds.), *Proceedings of the twelfth language resources and evaluation conference* (pp. 4478–4485). Marseille, France: European Language Resources Association. Retrieved from <https://aclanthology.org/2020.lrec-1.551>
- Leon, P. G., Ur, B., Wang, Y., Sleeper, M., Balebako, R., Shay, R., ... Cranor, L. F. (2013). What matters to users? factors that affect users’ willingness to share information with online advertisers. In *Proceedings of the ninth symposium on usable privacy and security* (pp. 1–12).
- Libert, T. (2018). An automated approach to auditing disclosure of third-party data collection in website privacy policies. In *Proceedings of the 2018 world wide web conference* (pp. 207–216).

- Lin, J., Liu, B., Sadeh, N., & Hong, J. I. (2014). Modeling users’ mobile app privacy preferences: Restoring usability in a sea of permission settings. In *10th symposium on usable privacy and security ({SOUPS} 2014)* (pp. 199–212).
- Linden, T., Khandelwal, R., Harkous, H., & Fawaz, K. (2020). The privacy policy landscape after the gdpr. *Proceedings on Privacy Enhancing Technologies*, 1, 47–64.
- Liu, F., Fella, N. L., & Liao, K. (2016). Modeling language vagueness in privacy policies using deep neural networks. In *2016 aaai fall symposium series*.
- Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., ... Stoyanov, V. (2019). Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*.
- Lloyd, S. (1982). Least squares quantization in pcm. *IEEE transactions on information theory*, 28(2), 129–137.
- Lu, J., & Li, D. (2010). Estimating deep web data source size by capture–recapture method. *Information retrieval*, 13(1), 70–95.
- Lui, M., & Baldwin, T. (2012a). langid. py: An off-the-shelf language identification tool. In *Proceedings of the acl 2012 system demonstrations* (pp. 25–30).
- Lui, M., & Baldwin, T. (2012b, July). langid.py: An off-the-shelf language identification tool. In *Proceedings of the ACL 2012 system demonstrations* (pp. 25–30). Jeju Island, Korea: Association for Computational Linguistics. Retrieved from <https://www.aclweb.org/anthology/P12-3005>
- Lutz, C., & Strathoff, P. (2014). Privacy concerns and online behavior—not so paradoxical after all? viewing the privacy paradox through different theoretical lenses. *Viewing the Privacy Paradox Through Different Theoretical Lenses (April 15, 2014)*.
- Malik, V., Sanjay, R., Nigam, S. K., Ghosh, K., Guha, S. K., Bhattacharya, A., & Modi, A. (2021, August). ILDC for CJPE: Indian legal documents corpus for court judgment prediction and explanation. In C. Zong, F. Xia, W. Li, & R. Navigli (Eds.), *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (volume 1: Long papers)* (pp. 4046–4062). Online: Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2021.acl-long.313> doi: 10.18653/v1/2021.acl-long.313
- Manku, G. S., Jain, A., & Das Sarma, A. (2007). Detecting near-duplicates for web crawling. In *Proceedings of the 16th international conference on world wide web* (pp. 141–150).
- McDonald, A. M., & Cranor, L. F. (2008). The cost of reading privacy policies. *Isjlp*, 4, 543.

- Meiselwitz, G. (2013). Readability assessment of policies and procedures of social networking sites. In *International conference on online communities and social computing* (pp. 67–75).
- Mhaidli, A., Fidan, S., Doan, A., Herakovic, G., Srinath, M., Matheson, L., ... Schaub, F. (2023). Researchers’ experiences in analyzing privacy policies: Challenges and opportunities. *Proceedings on Privacy Enhancing Technologies*.
- Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. In *Advances in neural information processing systems* (pp. 3111–3119).
- Mugan, J., Sharma, T., & Sadeh, N. (2011). Understandable learning of privacy preferences through default personas and suggestions. *Institute for Software Research Technical Report CMU-ISR-11-112. Carnegie Mellon University, Pittsburgh, PA*.
- Mumcuoğlu, E., Öztürk, C. E., Ozaktas, H. M., & Koç, A. (2021, September). Natural language processing in law: Prediction of outcomes in the higher courts of turkey. *Inf. Process. Manage.*, 58(5). Retrieved from <https://doi.org/10.1016/j.ipm.2021.102684> doi: 10.1016/j.ipm.2021.102684
- Nejad, N. M., Graux, D., & Collarana, D. (2019). Towards measuring risk factors in privacy policies. In *Proceedings of the first workshop on ai in the administrative state*.
- Nejad, N. M., Jabat, P., Nedelchev, R., Scerri, S., & Graux, D. (2020). Establishing a strong baseline for privacy policy classification. In *Ifip international conference on ict systems security and privacy protection*.
- Nejad, N. M., Scerri, S., & Lehmann, J. (2018). Knight: Mapping privacy policies to gdpr. In *European knowledge acquisition workshop* (pp. 258–272).
- Nokhbeh Zaeem, R., & Barber, K. S. (2021). A large publicly available corpus of website privacy policies based on dmoz. In *Proceedings of the eleventh acm conference on data and application security and privacy* (pp. 143–148).
- Norberg, P. A., Horne, D. R., & Horne, D. A. (2007). The privacy paradox: Personal information disclosure intentions versus behaviors. *Journal of consumer affairs*, 41(1), 100–126.
- Obar, J. A., & Oeldorf-Hirsch, A. (2018). The biggest lie on the internet: Ignoring the privacy policies and terms of service policies of social networking services. *Information, Communication & Society*, 1–20.
- O’Neil, D. (2001). Analysis of internet users’ level of online privacy concerns. *Social Science Computer Review*, 19(1), 17–31.

- Phelan, C., Lampe, C., & Resnick, P. (2016). It’s creepy, but it doesn’t bother me. In *Proceedings of the 2016 chi conference on human factors in computing systems* (pp. 5240–5251).
- Pollach, I. (2007). What’s wrong with online privacy policies? *Communications of the ACM*, 50(9), 103–108.
- Pollock, K. H., Nichols, J. D., Brownie, C., & Hines, J. E. (1990). Statistical inference for capture-recapture experiments. *Wildlife monographs*, 3–97.
- Ramanath, R., Liu, F., Sadeh, N., & Smith, N. A. (2014). Unsupervised alignment of privacy policies using hidden markov models. In *Proceedings of the 52nd annual meeting of the association for computational linguistics (volume 2: Short papers)* (pp. 605–610).
- Rao, A., Schaub, F., Sadeh, N., Acquisti, A., & Kang, R. (2016, June). Expecting the unexpected: Understanding mismatched privacy expectations online. In *Twelfth symposium on usable privacy and security (SOUPS 2016)* (pp. 77–96). Denver, CO: USENIX Association. Retrieved from <https://www.usenix.org/conference/soups2016/technical-sessions/presentation/rao>
- Ravichander, A., Black, A. W., Wilson, S., Norton, T., & Sadeh, N. (2019). Question answering for privacy policies: Combining computational and legal perspectives. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (emnlp-ijcnlp)* (pp. 4949–4959).
- Reimers, N., & Gurevych, I. (2019). Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (emnlp-ijcnlp)* (pp. 3982–3992).
- Robertson, S. E., Walker, S., Beaulieu, M., & Willett, P. (1999). Okapi at trec-7: automatic ad hoc, filtering, vlc and interactive track. *Nist Special Publication SP(500)*, 253–264.
- Rodrigues, R., Wright, D., & Wadhwa, K. (2013). Developing a privacy seal scheme (that works). *International Data Privacy Law*, 3(2), 100–116. doi: 10.1093/idpl/ips037
- Roy, H., Borchardt, J., McGowan, I., Stout, J., & Azmayesh, S. (2012). Terms of service; didn’t read. *Web Page, June*. URL <https://tosdr.org>.
- Rudolph, M., Feth, D., & Polst, S. (2018). Why users ignore privacy policies—a survey and intention model for explaining user privacy behavior. In *International conference on human-computer interaction* (pp. 587–598).

- Sarne, D., Schler, J., Singer, A., Sela, A., & Bar Siman Tov, I. (2019). Unsupervised topic extraction from privacy policies. In *Companion proceedings of the 2019 world wide web conference* (pp. 563–568).
- Sathyendra, K. M., Schaub, F., Wilson, S., & Sadeh, N. (2016). Automatic extraction of opt-out choices from privacy policies. In *2016 aaai fall symposium series*.
- Sathyendra, K. M., Wilson, S., Schaub, F., Zimmeck, S., & Sadeh, N. (2017). Identifying the provision of choices in privacy policy text. In *Proceedings of the 2017 conference on empirical methods in natural language processing* (pp. 2774–2779).
- Schaub, F., Balebako, R., Durity, A. L., & Cranor, L. F. (2015). A design space for effective privacy notices. In *Eleventh symposium on usable privacy and security ({SOUPS} 2015)* (pp. 1–17).
- Shah, A. D., Martinez, C., & Hemingway, H. (2012). The freetext matching algorithm: a computer program to extract diagnoses and causes of death from unstructured text in electronic health records. *BMC medical informatics and decision making*, 12(1), 1–13.
- Sharma, E., Li, C., & Wang, L. (2019, July). BIGPATENT: A large-scale dataset for abstractive and coherent summarization. In A. Korhonen, D. Traum, & L. Màrquez (Eds.), *Proceedings of the 57th annual meeting of the association for computational linguistics* (pp. 2204–2213). Florence, Italy: Association for Computational Linguistics. Retrieved from <https://aclanthology.org/P19-1212> doi: 10.18653/v1/P19-1212
- Shukla, A., Bhattacharya, P., Poddar, S., Mukherjee, R., Ghosh, K., Goyal, P., & Ghosh, S. (2022, November). Legal case document summarization: Extractive and abstractive methods and their evaluation. In Y. He, H. Ji, S. Li, Y. Liu, & C.-H. Chang (Eds.), *Proceedings of the 2nd conference of the asia-pacific chapter of the association for computational linguistics and the 12th international joint conference on natural language processing (volume 1: Long papers)* (pp. 1048–1064). Online only: Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2022.aacl-main.77>
- Sloan, R. H., & Warner, R. (2014). Beyond notice and choice: Privacy, norms, and consent. *J. High Tech. L.*, 14, 370.
- Smith, D. A. (2002). Detecting events with date and place information in unstructured text. In *Proceedings of the 2nd acm/ieee-cs joint conference on digital libraries* (pp. 191–196).
- Spiekermann, S., Grossklags, J., & Berendt, B. (2001). E-privacy in 2nd generation e-commerce: privacy preferences versus actual behavior. In *Proceedings of the 3rd acm conference on electronic commerce* (pp. 38–47).

- Srinath, M., Sundareswara, S. N., Giles, C. L., & Wilson, S. (2021). Privaseer: A privacy policy search engine. In *International conference on web engineering* (pp. 286–301).
- Srinath, M., Wilson, S., & Giles, C. L. (2020). Privacy at scale: Introducing the privaseer corpus of web privacy policies. *arXiv preprint arXiv:2004.11131*.
- Srinath, M., Wilson, S., & Giles, C. L. (2021). Privacy at scale: Introducing the privaseer corpus of web privacy policies. In *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (volume 1: Long papers)* (pp. 6829–6839).
- Story, P., Zimmeck, S., Ravichander, A., Smullen, D., Wang, Z., Reidenberg, J., ... Sadeh, N. (2019). Natural language processing for mobile app privacy compliance. In *Aaai spring symposium on privacy-enhancing artificial intelligence and language technologies*.
- Sundareswara, S. N., Srinath, M., Wilson, S., & Giles, C. L. (2021). A large-scale exploration of terms of service documents on the web. In *Proceedings of the 21st acm symposium on document engineering*. New York, NY, USA: Association for Computing Machinery. Retrieved from <https://doi.org/10.1145/3469096.3474940> doi: 10.1145/3469096.3474940
- Sundareswara, S. N., Wilson, S., Srinath, M., & Giles, C. L. (2020). Privacy not found: A study of the availability of privacy policies on the web. In *Sixteenth symposium on usable privacy and security (SOUPS 2020)*. USENIX Association.
- Sunyaev, A., Dehling, T., Taylor, P., & Mandl, K. (2014, 08). Availability and quality of mobile health app privacy policies. *Journal of the American Medical Informatics Association*, 1–4. doi: 10.1136/amiajnl-2013-002605
- Supervisor, F. . E. D. P. (2018). What to expect when we inspect. Retrieved from https://edps.europa.eu/sites/edp/files/publication/18-11-21_factsheet_inspections_en.pdf
- Verisign. (n.d.). *Verisign q1 2020 domain name industry brief*. Retrieved from <https://blog.verisign.com/domain-names/verisign-q1-2020-domain-name-industry-brief-internet-grows-to-366-8-million-domain-name-registrations-in-the-first-quarter-of-2020/>
- Vu, K.-P. L., Chambers, V., Creekmur, B., Cho, D., & Proctor, R. W. (2010). Influence of the privacy bird® user agent on user trust of different web sites. *Computers in industry*, 61(4), 311–317.
- Wallach, H. M. (2006). Topic modeling: beyond bag-of-words. In *Proceedings of the 23rd international conference on machine learning* (pp. 977–984).

- Weaver, R., & Collins, M. P. (2007). Fishing for phishes: Applying capture-recapture methods to estimate phishing populations. In *Proceedings of the anti-phishing working groups 2nd annual ecrime researchers summit* (pp. 14–25).
- Weninger, T., Palacios, R., Crescenzi, V., Gottron, T., & Merialdo, P. (2016). Web content extraction: a metaanalysis of its past and thoughts on its future. *ACM SIGKDD Explorations Newsletter*, 17(2), 17–23.
- Wikipedia contributors. (2020). *Lorem ipsum — Wikipedia, the free encyclopedia*. https://en.wikipedia.org/w/index.php?title=Lorem_ipsum&oldid=953944429. ([Online; accessed 12-May-2020])
- Wilson, S., Schaub, F., Dara, A. A., Liu, F., Cherivirala, S., Leon, P. G., ... Sadeh, N. (2016). The creation and analysis of a website privacy policy corpus. In *Proceedings of the 54th annual meeting of the association for computational linguistics* (pp. 1330–1340).
- Wolf, T., Chaumond, J., Debut, L., Sanh, V., Delangue, C., Moi, A., ... others (2020). Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 conference on empirical methods in natural language processing: System demonstrations* (pp. 38–45). doi: 10.18653/v1/2020.emnlp-demos.6
- Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., ... Rush, A. M. (2019). Huggingface’s transformers: State-of-the-art natural language processing. *ArXiv*, *abs/1910.03771*.
- Xiao, C., Zhong, H., Guo, Z., Tu, C., Liu, Z., Sun, M., ... others (2018). Cail2018: A large-scale legal dataset for judgment prediction. *arXiv preprint arXiv:1807.02478*.
- Young, A. L., & Quan-Haase, A. (2013). Privacy protection strategies on facebook: The internet privacy paradox revisited. *Information, Communication & Society*, 16(4), 479–500.
- Zaeem, R. N., German, R. L., & Barber, K. S. (2018). Privacycheck: Automatic summarization of privacy policies using data mining. *ACM Transactions on Internet Technology (TOIT)*, 18(4), 1–18. doi: 10.1145/3127519
- Zhang, S., & Sadeh, N. (2023). Do privacy labels answer users’ privacy questions. In *Network and distributed system security symposium*.
- Zheng, L., Guha, N., Anderson, B. R., Henderson, P., & Ho, D. E. (2021). When does pretraining help? assessing self-supervised learning for law and the casehold dataset of 53,000+ legal holdings. In *Proceedings of the eighteenth international conference on artificial intelligence and law* (pp. 159–168).
- Zimmeck, S., & Bellovin, S. M. (2014). Privee: An architecture for automatically analyzing web privacy policies. In *23rd {USENIX} security symposium ({USENIX} security 14)* (pp. 1–16).

Zimmeck, S., Story, P., Smullen, D., Ravichander, A., Wang, Z., Reidenberg, J., ...
Sadeh, N. (2019). Maps: Scaling privacy compliance analysis to a million apps.
Proceedings on Privacy Enhancing Technologies, 2019(3), 66–86.

Mukund Srinath

Education

Pennsylvania State University

PhD, Informatics

Co-supervised: C. Lee Giles and Shomir Wilson

August 2019 - December 2024

GPA: 3.85

R. V. College of Engineering, India

Bachelor of Engineering, Electrical Engineering

August, 2011 - May, 2015

GPA: 8.94

Experience

Human Language Technologies Lab

Research Assistant

State College, PA

August, 2019 - Present

- Pretrained and fine-tuned large language model (LLM) for privacy policy documents. [[HuggingFace](#)]
- Designed and developed, a retrieval augmented generation (RAG) pipeline for a privacy policy search engine. [[PrivaSeer](#)]
- Experimented with instruction tuning and reinforcement learning from human feedback (RLHF) for privacy policy LLM.
- Achieved SoTA results on semantic similarity, named entity recognition (NER), and question-answering (QA) tasks.

Intelligent Information Systems Lab

Machine Learning Researcher

State College, PA

August, 2019 - Present

- Engineered prompts to identify biases by jail-breaking GenAI models using chain-of-thought (CoT) prompting.
- Designed approach to measure disability bias in sentiment analysis and toxicity detection models.
- Identified gaps in GenAI models' definition of hallucination, creating a taxonomy for better evaluation and identification.

Trustpage

Machine Learning Intern

Detroit, MI

May, 2022 - August, 2022

- Designed evaluation metrics to track the performance of a suite of machine learning models.
- Implemented and deployed a question similarity model on Google Cloud (GCP), improving performance by 4%.
- Applied feature engineering and synthetic data generation techniques on low-resource tasks.

Skills

Machine Learning Skills: PyTorch, Tensorflow, sklearn. Experience with multi-GPU processing.

Development Skills: Python, R, Java, C++, Elasticsearch, Django, SQL, NoSQL, Git and multiple OS.

Awards and Honors

- [Liberatory Tech Fellow](#), with Logic(s) Magazine and Columbia University 2024
- **Out-of-the-box winner**, [Bias-a-thon](#), for identifying novel biases in LLMs 2023
- **Outstanding Paper**, EMNLP 2023
- **Best Student Paper**, DocEng 2023
- **Best Short Paper**, TrustNLP Workshop, ACL 2023
- **Best Student Paper**, PETS 2023

Selected Publications

[Automated Detection and Analysis of Data Practices Using A Real-World Corpus](#), **ACL 2024**

[The Sentiment Problem: A Critical Survey towards Deconstructing Sentiment Analysis](#), **EMNLP 2023** [**Outstanding Paper**]

[Automated Ableism: An Exploration of Explicit Disability Biases in Sentiment and Toxicity Analysis Models](#) **TrustNLP Workshop, ACL, 2023** [**Best Paper**]

[A Study of Implicit Language Model Bias Against People With Disabilities](#) **COLING, 2022**

[Privacy at Scale: Introducing the PrivaSeer Corpus of Web Privacy Policies](#). **ACL, 2021**